



Workshop Manual

Crafter 2017 ➤

TGE 2017 ➤

Communication

Edition 03.2018



List of Workshop Manual Repair Groups

Repair Group

00 - Technical data

91 - Communication

Technical information should always be available to the foremen and mechanics, because their careful and constant adherence to the instructions is essential to ensure vehicle road-worthiness and safety. In addition, the normal basic safety precautions for working on motor vehicles must, as a matter of course, be observed.



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00 – Technical data

1 Repair instructions

(VRL011408; Edition 03.2018)

⇒ [“1.1 General repair instructions”, page 1](#)

⇒ [“1.2 Routing and attachment of lines”, page 1](#)

⇒ [“1.3 Notes on repair of CAN bus wiring”, page 1](#)

⇒ [“1.4 General information”, page 1](#)

1.1 General repair instructions

- Repairing wiring harnesses and electrical connectors ⇒ Electrical system, General information; Rep. gr. 97 ; Electrical connectors; Repairing wiring harnesses and electrical connectors .
- Releasing and dismantling connector housings ⇒ Electrical system; General information; Rep. gr. 97 ; Releasing and dismantling connector housings .
- Cleaning contact surfaces ⇒ Electrical system, General information; Rep. gr. 97 ; Cleaning contact surfaces .
- Repairing aerial wires ⇒ Electrical system; General information; Rep. gr. 97 ; Repairing aerial wires .
- Repairing fibre optic cables ⇒ Electrical system; General information; Rep. gr. 97 ; Fibre optic cables .
- Vehicle diagnostic, testing and information systems ⇒ Electrical system, General information; Rep. gr. 97 ; Vehicle diagnostic, testing and information systems

1.2 Routing and attachment of lines

- ◆ Mark lines prior to removal to prevent them from being interchanged and to ensure that they are fitted in their original positions. This applies for fuel, hydraulic and vacuum lines as well as lines for activated charcoal filter system and electrical wiring. Where necessary, make sketches or take photographs.
- ◆ Any wiring fasteners which have been released during removal must be of the same type and must be attached at the same locations when installing to prevent any rattling or damage.

1.3 Notes on repair of CAN bus wiring

- Repairing CAN bus wiring ⇒ Electrical system, General information; Rep. gr. 97 ; Wiring harness and connector repair; Wiring harness and connector repair .

1.4 General information

⇒ [“1.4.1 Transmitting power and aerial fitting locations”, page 1](#)

1.4.1 Transmitting power and aerial fitting locations



Note

Installation of aerials on bumpers is now prohibited by law. This improves pedestrian protection during accidents.



Designation	Pmax (watts)	Prescribed aerial fitting locations
Short wave < 54 MHz	100 (PEAK) ¹⁾	All locations on roof
4 m frequency band	20 (effective) ²⁾	All locations on roof
2 m frequency band	20 (effective) ²⁾	All locations on roof
2 m frequency band	50 (effective) ²⁾	All locations on roof
70 cm	50 (effective) ²⁾	All locations on roof
23 cm	20 (effective) ²⁾	All locations on roof
TETRA/ TETRAPOL	25 (effective) ²⁾	All locations on roof
D network GSM 900	10 (Peak) ¹⁾	All locations on roof
E network GSM 1800 UMTS	10 (Peak) ¹⁾	All locations on roof

1) PEAK = peak envelope power (max. carrier power)

2) effective = effective transmitting power



Note

- ◆ *Deviations from these specifications (aerial fitting location, frequency, power) are permitted only in specially justified exceptions following a single-case test performed by the EMC Centre of the manufacturer.*
- ◆ *EMC = electromagnetic compatibility*



91 – Communication

1 Infotainment system

⇒ [“1.1 Layout - infotainment system”, page 3](#)

⇒ [“1.2 Overview of fitting locations - Infotainment system”, page 8](#)

⇒ [“1.3 Removing and installing infotainment system display”, page 10](#)

⇒ [“1.4 Removing and installing control unit 1 for information electronics J794”, page 11](#)

1.1 Layout - infotainment system

Components of infotainment system, Composition Audio version
⇒ [page 3](#)

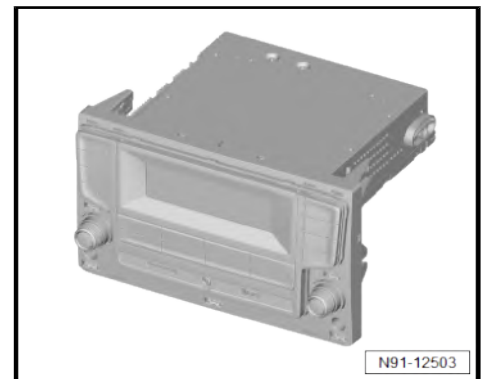
Components of Infotainment system, Composition Media and Discover Media versions ⇒ [page 5](#)

Components of infotainment system, Composition Audio version

- ◆ Display unit for front information display and operating unit control unit - J685-

Features of infotainment system Composition Audio

- ◆ Monochrome display
- ◆ 2 or 4 x 20 watt output power
- ◆ SD memory card reader with 1 SD card slot
- ◆ AUX IN on front of unit via jack connector
- ◆ Front USB connection
- ◆ RDS FM/AM radio
- ◆ Bluetooth hands-free system



Note

- ◆ *To deal with complaints it is paramount that the function and operation of the Infotainment system is understood.*
- ◆ *Additional information can be found in the ⇒ Operating manual.*
- ◆ *Other systems in the vehicle are connected via CAN bus (infotainment) and data bus diagnostic interface - J533-.*
- ◆ *Fault finding is performed using ⇒ Vehicle diagnostic tester.*

Infotainment system overview for Composition Audio



1 - Roof aerial - RX5-

- ❑ Removing and installing
⇒ [page 19](#)

2 - Interior microphone - R74-

- ❑ Removing and installing
⇒ [page 35](#)

3 - Display unit for control device of front display and information control panel - J685-

- ❑ Removing and installing
⇒ [page 29](#)

4 - Mid-range loudspeaker, front right - R104-

- ❑ Wide band loudspeakers
- ❑ Removing and installing
⇒ [page 13](#)

5 - Rear right mid-range and bass loudspeaker - R160-

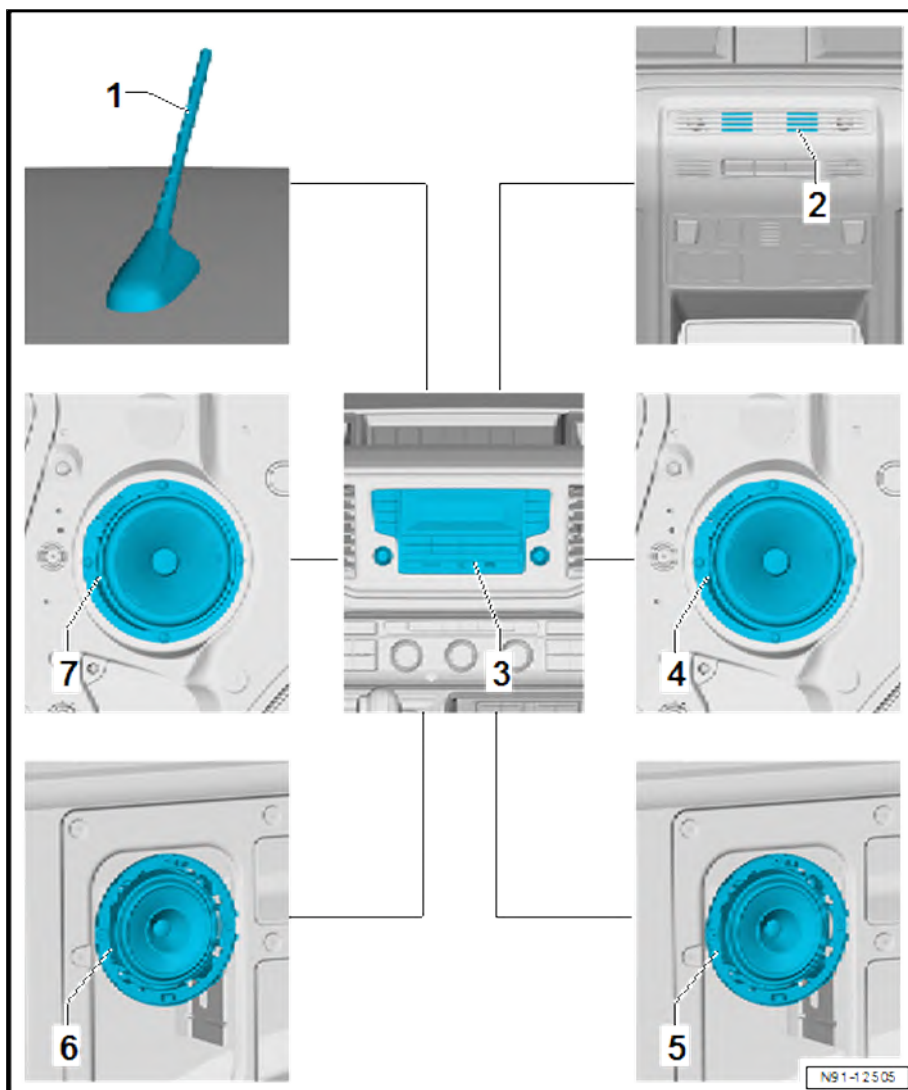
- ❑ Wide band loudspeakers
- ❑ Removing and installing
⇒ [page 13](#)

6 - Rear left mid-range and bass loudspeaker - R159-

- ❑ Wide band loudspeakers
- ❑ Removing and installing
⇒ [page 13](#)

7 - Mid-range loudspeaker, front left - R103-

- ❑ Wide band loudspeakers
- ❑ Removing and installing ⇒ [page 13](#)





Components of Infotainment system, Composition Media and Discover Media versions

- ◆ Control unit 1 for information electronics - J794- / display unit for front information display and operating unit control unit - J685-

Features of Composition Media infotainment system

- ◆ 8-inch multicolour touch screen (approach detection)
- ◆ 4 front loudspeakers
- ◆ Optional 2 loudspeakers in luggage/passenger compartment
- ◆ 4 x 20 watt output power
- ◆ RDS FM/AM radio
- ◆ Integrated diversity function
- ◆ SD memory card reader
- ◆ CD drive mp3/WMA
- ◆ Connections for external multimedia devices
- ◆ Connections for external audio sources
- ◆ USB connection
- ◆ Connection for iPod and iPhone
- ◆ Support for multifunction steering wheel
- ◆ Support for vehicle menu
- ◆ Self-diagnosis, including loudspeaker diagnosis
- ◆ Bluetooth hands-free system
- ◆ Speed dependent volume control (GALA)
- ◆ Connection for reversing camera
- ◆ Voice control

Features of infotainment system; Composition Discover Media

- ◆ 8-inch multicolour touch screen (approach detection)
- ◆ 4 front loudspeakers
- ◆ 4 x 20 watt output power
- ◆ Optional 2 loudspeakers in luggage/passenger compartment
- ◆ RDS FM/AM radio
- ◆ Digital radio reception (optional)
- ◆ Integrated diversity function
- ◆ SD memory card reader
- ◆ CD drive mp3/WMA
- ◆ Connection for external multimedia devices
- ◆ Connection for external audio sources
- ◆ USB connection
- ◆ Connection for iPod and iPhone
- ◆ Support for multifunction steering wheel
- ◆ Support for vehicle menu
- ◆ Self-diagnosis, including loudspeaker diagnosis





- ◆ Bluetooth hands-free system
- ◆ Speed dependent volume control (GALA)
- ◆ FBAS connection for reversing camera
- ◆ Voice control
- ◆ Navigation function
- ◆ Lifetime maps update service
- ◆ Car-Net "Guide & Inform"



Note

- ◆ *To deal with complaints it is paramount that the function and operation of the Infotainment system is understood.*
- ◆ *Additional information can be found in the ⇒ Operating manual .*
- ◆ *Other systems in the vehicle are connected via CAN bus (infotainment) and data bus diagnostic interface - J533- .*
- ◆ *Fault finding is performed using ⇒ Vehicle diagnostic tester.*

Infotainment system overview, Composition Media and Discover Media



1 - Roof aerial - RX5-

- ❑ Removing and installing
⇒ [page 19](#)

2 - Multifunction buttons on left in steering wheel - E440-

- ❑ Removing and installing
⇒ [page 55](#)

3 - Multifunction buttons on right in steering wheel - E441-

- ❑ Removing and installing
⇒ [page 55](#)

4 - Interior microphone - R74-

- ❑ Removing and installing
⇒ [page 35](#)

5 - Information electronics control unit 1 - J794-

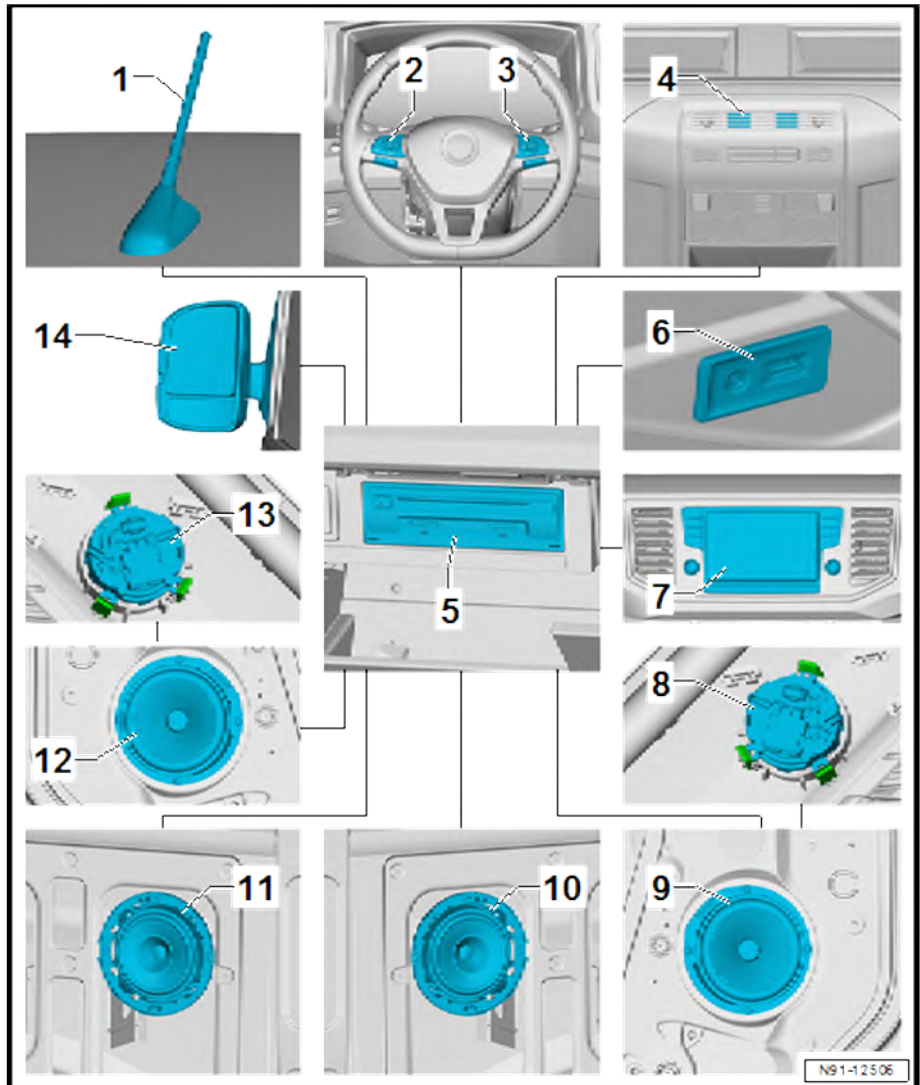
- ❑ Removing and installing
⇒ [page 11](#)
- ❑ Pin assignment ⇒ Current flow diagrams, Electrical fault finding and Fitting locations

6 - Connection for external multimedia devices

- ❑ Removing and installing
⇒ [page 60](#)

7 - Operating and display unit for multimedia system - R238-

- ❑ Removing and installing
⇒ [page 10](#)
- ❑ Pin assignment ⇒ Current flow diagrams, Electrical fault finding and Fitting locations



8 - Front right treble loudspeaker - R22-

- ❑ Removing and installing ⇒ [page 15](#)

9 - Bass loudspeaker, front right - R23-

- ❑ Removing and installing ⇒ [page 17](#)

10 - Bass loudspeaker, rear right - R17-

- ❑ Removing and installing ⇒ [page 15](#)

11 - Bass loudspeaker, rear left - R15-

- ❑ Removing and installing ⇒ [page 15](#)

12 - Bass loudspeaker, front left - R21-

- ❑ Removing and installing ⇒ [page 17](#)

13 - Front left treble loudspeaker - R20-

- ❑ Removing and installing ⇒ [page 15](#)

14 - Left exterior mirror

- ❑ With radio aerial - R248- for AM/FM
- ❑ With GSM aerial - R205- for telephone
- ❑ Removing and installing aerial module ⇒ [page 24](#)

1.2 Overview of fitting locations - Infotainment system

Overview of fitting locations - Infotainment system, Infotainment system Composition Audio ➔ [page 8](#)

Overview of fitting locations - Infotainment system, Infotainment system Composition Media and Discover Media ➔ [page 8](#)

Overview of fitting locations - Infotainment system, Infotainment system Composition Audio

1 - Rear right mid-range and bass loudspeaker - R160-

- ☐ in side panel trim
- ☐ Removing and installing ➔ [page 15](#)

2 - Rear left mid-range and bass loudspeaker - R159-

- ☐ in side panel trim
- ☐ Removing and installing ➔ [page 15](#)

3 - Display unit for control device of front display and information control panel - J685-

- ☐ Removing and installing ➔ [page 29](#)

4 - Mid-range loudspeaker, front left - R103-

- ☐ in front door trim
- ☐ Removing and installing ➔ [page 17](#)

5 - Mid-range loudspeaker, front right - R104-

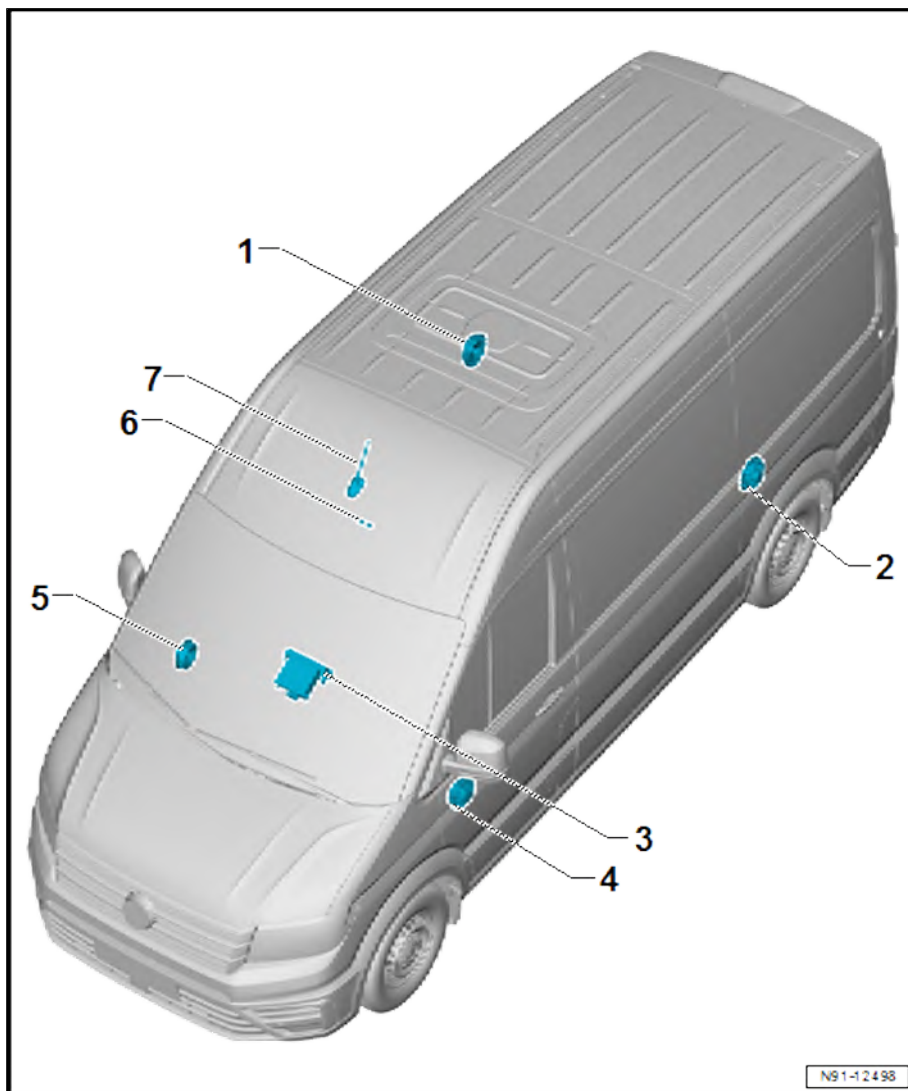
- ☐ in front door trim
- ☐ Removing and installing ➔ [page 17](#)

6 - Interior microphone - R74-

- ☐ Removing and installing ➔ [page 35](#)

7 - Roof aerial - RX5-

- ☐ Removing and installing ➔ [page 19](#)



Overview of fitting locations - Infotainment system, Infotainment system Composition Media and Discover Media



1 - Left exterior mirror with aerial system

- ☐ Removing and installing
⇒ [page 19](#)

2 - Front left treble loudspeaker - R20-

- ☐ In front A-pillar
- ☐ Removing and installing
⇒ [page 15](#)

3 - Connection for external multimedia devices

- ☐ Removing and installing
⇒ [page 60](#)

4 - Multifunction buttons on left in steering wheel - E440-

- ☐ Removing and installing
⇒ [page 55](#)

5 - Multifunction buttons on right in steering wheel - E441-

- ☐ Removing and installing
⇒ [page 55](#)

6 - Operating and display unit for multimedia system - R238-

- ☐ Removing and installing
⇒ [page 10](#)
- ☐ Pin assignment ⇒ Current flow diagrams, Electrical fault finding and Fitting locations

7 - Interior microphone - R74-

- ☐ Removing and installing
⇒ [page 35](#)

8 - Roof aerial - RX5-

- ☐ Removing and installing ⇒ [page 19](#)

9 - Information electronics control unit 1 - J794-

- ☐ Removing and installing ⇒ [page 11](#)
- ☐ Pin assignment ⇒ Current flow diagrams, Electrical fault finding and Fitting locations

10 - Front right treble loudspeaker - R22-

- ☐ In front A-pillar
- ☐ Removing and installing ⇒ [page 15](#)

11 - Bass loudspeaker, front right - R23-

- ☐ in front door trim
- ☐ Removing and installing ⇒ [page 17](#)

12 - Bass loudspeaker, rear right - R17-

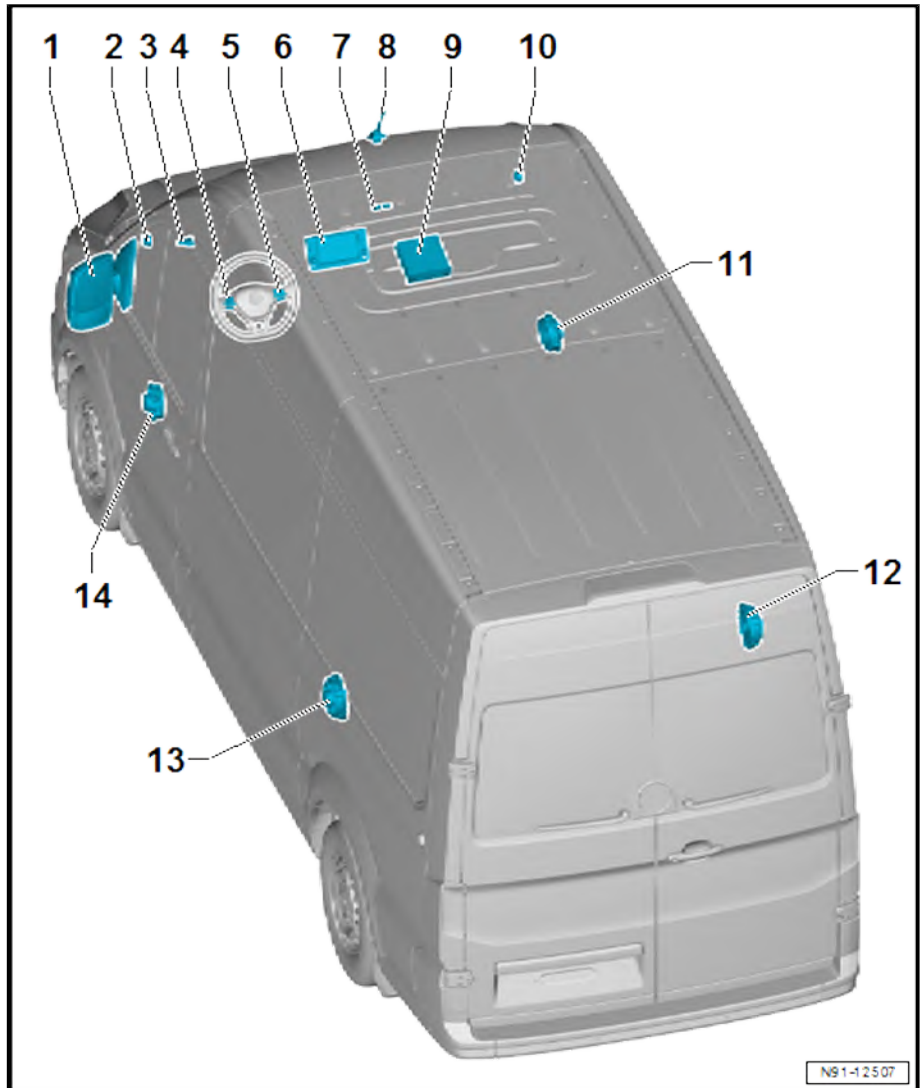
- ☐ in side panel trim
- ☐ Removing and installing ⇒ [page 15](#)

13 - Bass loudspeaker, rear left - R15-

- ☐ in side panel trim
- ☐ Removing and installing ⇒ [page 15](#)

14 - Bass loudspeaker, front left - R21-

- ☐ in front door trim





- ❑ Removing and installing ⇒ [page 17](#)

1.3 Removing and installing infotainment system display

Special tools and workshop equipment required

- ◆ Radio release tool - T10057-



Removing

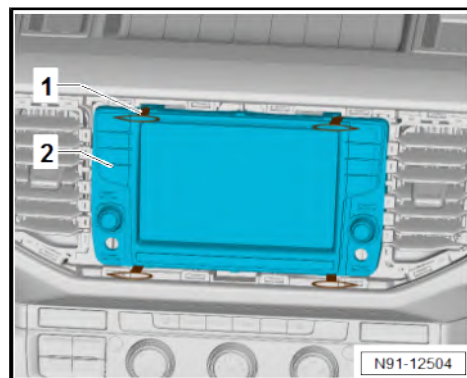
- Remove centre vent ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing dash panel vent .
- Insert radio release tool - T10057- into 4 release slots -1- of operating and display unit for multimedia system - R238- .



Note

The pointed projections on the grip rings of radio release tools - T10057- must point outwards.

- Release upper locking elements by pressing them in direction of travel.
- Release lower locking elements by pressing them in direction of travel.
- Pull out operating and display unit for multimedia system - R238- -2-.
- Separate electrical connectors.



Installing

Install in the reverse order of removal, observing the following:

When installing, do not press on the display and the control elements.

Operating and display unit for multimedia system - R238- must engage audibly when installing.



1.4 Removing and installing control unit 1 for information electronics - J794-

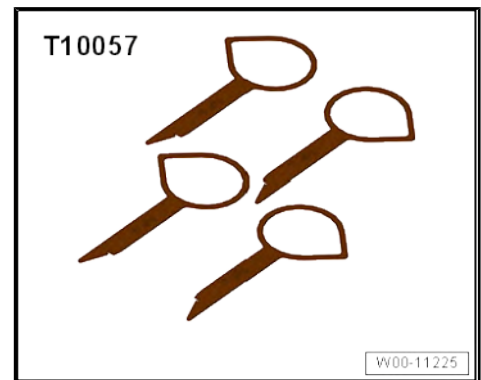
⇒ ["1.4.1 Removing and installing control unit 1 for information electronics J794", page 11](#)

⇒ ["1.4.2 Removing and installing line guide for control unit 1 for information electronics J794", page 12](#)

1.4.1 Removing and installing control unit 1 for information electronics - J794-

Special tools and workshop equipment required

- ◆ Radio release tool - T10057-



Removing

- Eject any data medium that is still in control unit 1 for information electronics - J794- .
- Switch off ignition and all electrical consumers.
- Keep the ignition key outside the vehicle to prevent the ignition from being switched on unintentionally.
- Open glove compartment.
- Insert radio release tools - T10057- -2- into slots of control unit 1 for information electronics - J794- -1-.



Note

The pointed projections on the grip rings of radio release tools - T10057- must point outwards.

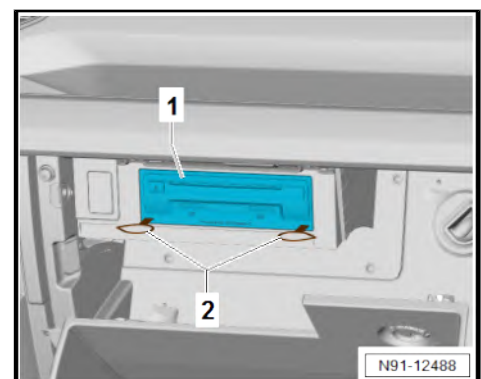
- Remove control unit 1 for information electronics - J794- -1- from slot by pulling on eyes of radio release tools - T10057- .
- Separate electrical connectors.
- Press catches on control unit 1 for information electronics - J794- , and pull out radio release tools - T10057- .

Installing

Install in the reverse order of removal, observing the following:

Control unit 1 for information electronics - J794- must engage audibly during installation.

If the control unit 1 for information electronics - J794- is renewed, start the respective function ⇒ Vehicle diagnostic tester.

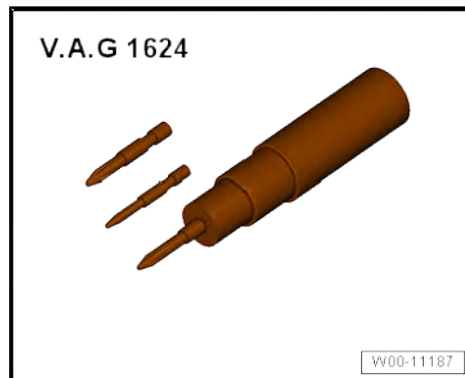




1.4.2 Removing and installing line guide for control unit 1 for information electronics - J794-

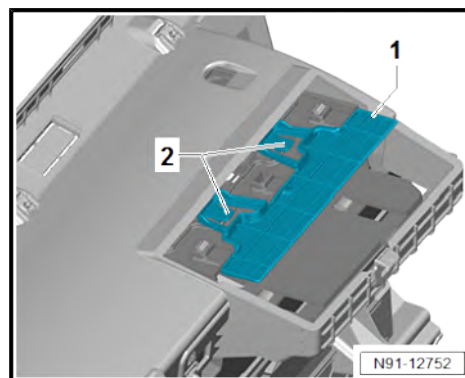
Special tools and workshop equipment required

- ◆ Torque screwdriver - V.A.G 1624-



Removing

- Remove glove compartment ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Removing and installing glove compartment .
- Release catches -2-. At the same time, pull out line guide -1-.



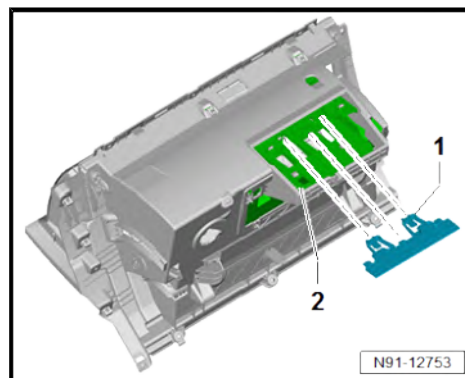
Installing

Install in the reverse order of removal, observing the following:

- Insert line guide -1- into guide -2-.

Specified torques

- ◆ Compartments/covers; Assembly overview - glove compartment ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Assembly overview - glove compartment .





2 Sound system

⇒ [“2.1 Components of sound system”, page 13](#)

⇒ [“2.2 Overview of fitting locations - sound system”, page 13](#)

⇒ [“2.3 Removing and installing front treble loudspeakers R20 / R22”, page 15](#)

⇒ [“2.4 Removing and installing rear bass loudspeakers R15 / R17”, page 15](#)

⇒ [“2.5 Removing and installing front bass loudspeakers R21 / R23”, page 17](#)

⇒ [“2.6 Removing and installing loudspeaker for emergency call module R335”, page 18](#)

2.1 Components of sound system

Vehicles with 2 loudspeakers

The sound system in a vehicle with 2 loudspeakers comprises the following components:

- ◆ One wide band loudspeaker each in front left and right door

Vehicles with 4 loudspeakers

The sound system in a vehicle with 4 loudspeakers comprises the following components:

- ◆ One bass loudspeaker each in front left and right door
- ◆ One treble loudspeaker in each A-pillar trim on left and right

Vehicles with 6 loudspeakers

The sound system in a vehicle with 6 loudspeakers comprises the following components:

- ◆ One bass loudspeaker each in front left and right door
- ◆ One treble loudspeaker in each A-pillar trim on left and right
- ◆ One wide band loudspeaker each in rear left and right side panel trim

2.2 Overview of fitting locations - sound system

1 - Rear right loudspeaker

- ☐ in rear side trim panel
- ☐ In conjunction with the Composition Audio Infotainment system there is also a rear right mid-range and bass loudspeaker - R160- here
- ☐ In conjunction with all other Infotainment systems there is also a rear right bass loudspeaker - R17- here
- ☐ Removing and installing ⇒ [page 15](#)

2 - Rear left loudspeaker

- ☐ in rear side trim panel
- ☐ In conjunction with the Composition Audio Infotainment system there is also a rear left mid-range and bass loudspeaker - R159- here
- ☐ In conjunction with all other Infotainment systems there is also a rear left bass loudspeaker - R15- here
- ☐ Removing and installing ⇒ [page 15](#)

3 - Loudspeaker for emergency call module - R335-

- ☐ in trim in front of dash panel insert
- ☐ for Car-Net/emergency call function
- ☐ Removing and installing ⇒ [page 18](#)

4 - Front left treble loudspeaker - R20-

- ☐ in left A-pillar trim
- ☐ Not installed in conjunction with Composition Audio Infotainment system
- ☐ Removing and installing ⇒ [page 15](#)

5 - Front left loudspeaker

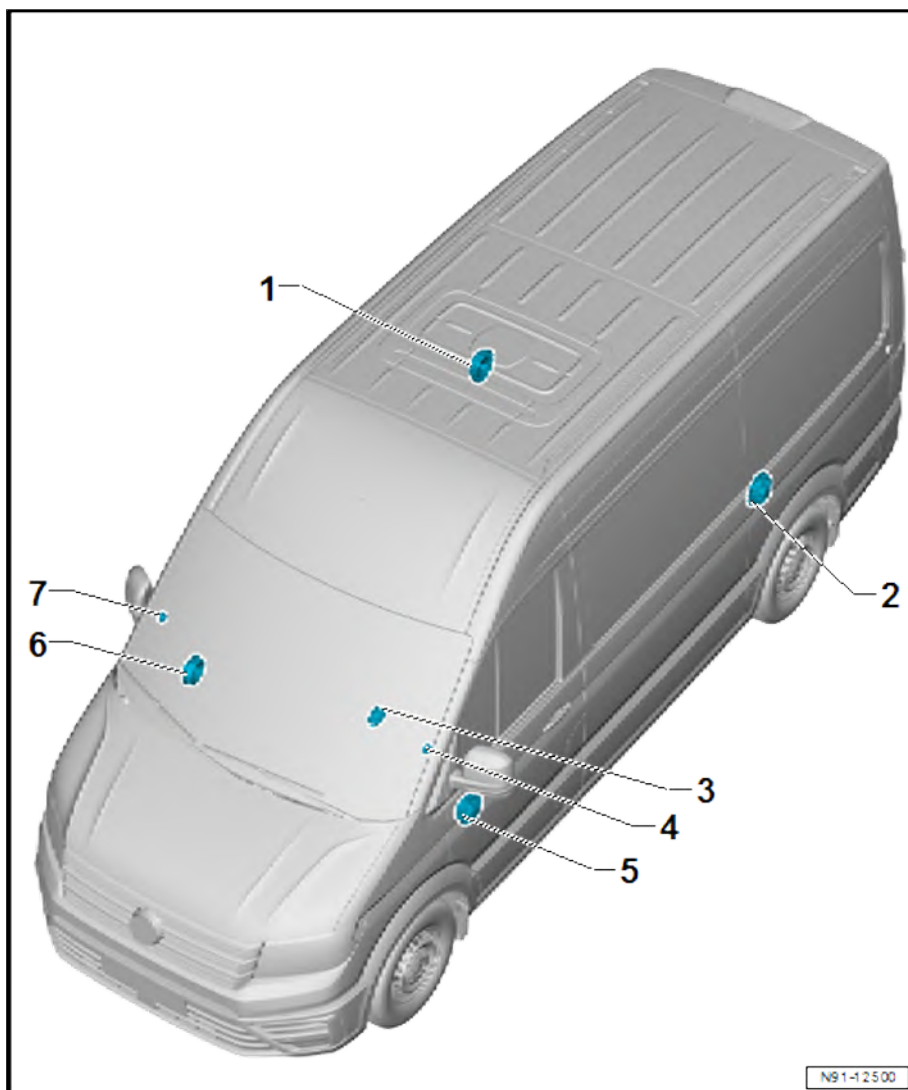
- ☐ in front door trim
- ☐ In conjunction with the Composition Audio Infotainment system there is also a front left mid-range loudspeaker - R103- here
- ☐ In conjunction with all other Infotainment systems there is also a front left bass loudspeaker - R21- here
- ☐ Removing and installing ⇒ [page 17](#)

6 - Front right loudspeaker

- ☐ in front door trim
- ☐ In conjunction with the Composition Audio Infotainment system there is also a front right mid-range loudspeaker - R104- here
- ☐ In conjunction with all other Infotainment systems there is also a front right bass loudspeaker - R23- here
- ☐ Removing and installing ⇒ [page 17](#)

7 - Front right treble loudspeaker - R22-

- ☐ in right A-pillar trim
- ☐ Not installed in conjunction with Composition Audio Infotainment system





❑ Removing and installing ⇒ [page 15](#)

2.3 Removing and installing front treble loudspeakers -R20- / -R22-

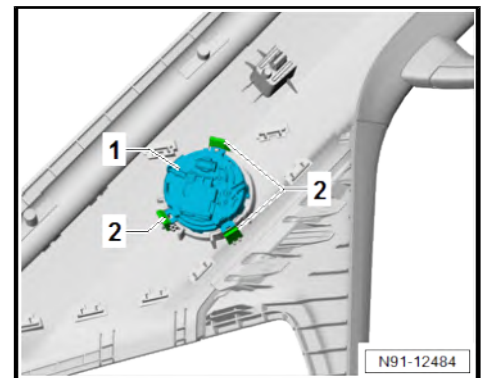
Removing

- Switch off ignition and all electrical consumers.
- Remove A-pillar trim ⇒ General body repairs, interior; Rep. gr. 70 ; Interior trim; Removing and installing A-pillar trim .
- Separate electrical connector.
- Push retaining tabs -2- and remove loudspeaker -1-.

Installing

Install in the reverse order of removal, observing the following:

Speaker -1- must be heard to engage in locking lugs -2- when installed.



2.4 Removing and installing rear bass loudspeakers -R15- / -R17-

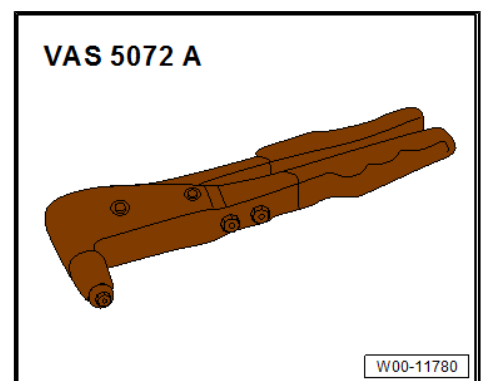


Note

For installation of the rear right mid-range and bass loudspeaker - R160- or the rear left mid-range and bass loudspeaker - R159- in this position, derive the removal and installation procedure from the following description.

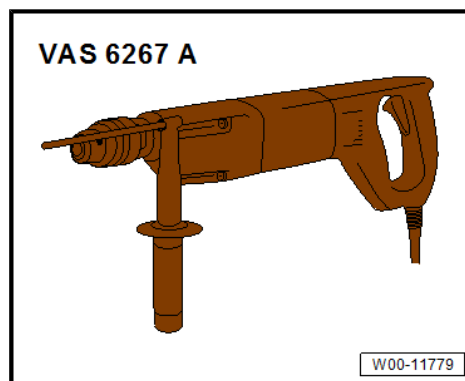
Special tools and workshop equipment required

- ◆ Pop rivet pliers - VAS 5072 A-





◆ Hand drill - VAS 6267 A-



- ◆ Drill bit with diameter smaller than 4.5 mm

Removing

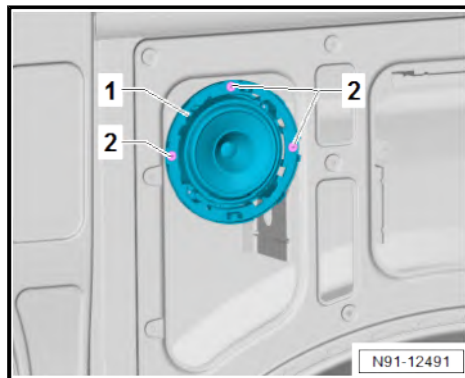
- Switch off ignition and all electrical consumers.
- Remove rear side trim ⇒ General body repairs, interior; Rep. gr. 70 ; Trims, interior; Removing and installing side trim .
- Separate electrical connector on loudspeaker -1-.
- Drill out pop rivets -2- using drill - VAS 6267 A- and a drill bit with a diameter smaller than 4.5 mm.
- Remove loudspeaker with bracket.
- Remove all swarf from the door.



Note

Should the paintwork become damaged, repair it immediately ⇒ General paint information; Rep. gr. 00 ; General notes; Paint structure of paintwork applied by customer service .

- Unscrew bolts -3- from bracket -1-.
- Remove loudspeaker -2- from bracket.



Installing

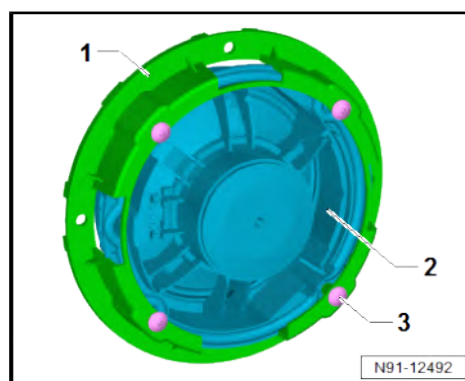
Install in the reverse order of removal, observing the following:



Note

Only secure the loudspeaker bracket to the body using the pre-scribed pop rivets ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

- Install pop rivets with pop rivet pliers - VAS 5072 A- .



Specified torques

- ◆ ⇒ [“2.2 Overview of fitting locations - sound system”, page 13](#)



2.5 Removing and installing front bass loudspeakers -R21- / -R23-

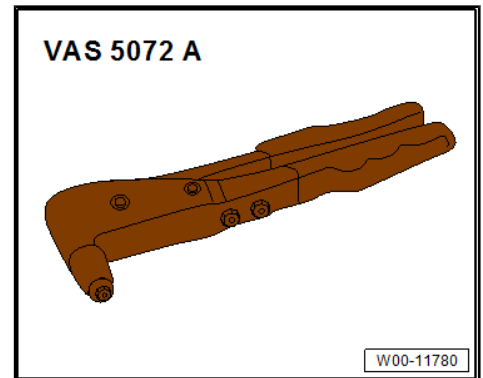


Note

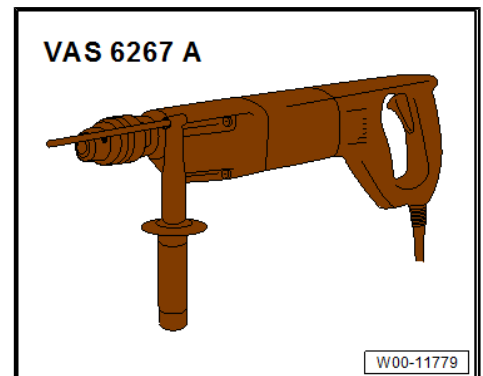
For installation of the front right mid-range loudspeaker - R104- or the front left mid-range loudspeaker - R103- in this position, derive the removal and installation procedure from the following description.

Special tools and workshop equipment required

- ◆ Pop rivet pliers - VAS 5072 A-



- ◆ Hand drill - VAS 6267 A-



- ◆ Drill bit with diameter smaller than 4.5 mm

Removing

- Switch off ignition and all electrical consumers.
- Remove front door trim panel ⇒ General body repairs, interior; Rep. gr. 70 ; Front door trim panels; Removing and installing front door trim panel .
- Separate electrical connector on loudspeaker.

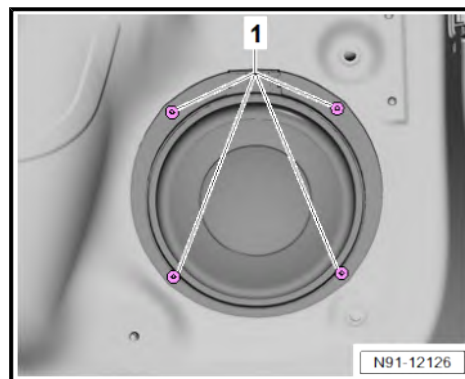


- Drill out pop rivets -1- using drill - VAS 6267 A- and a drill bit with a diameter smaller than 4.5 mm.
- Remove loudspeaker.
- Remove all swarf from the door.



Note

Should the paintwork become damaged, repair it immediately ⇒ General paint information; Rep. gr. 00 ; General notes; Paint structure of paintwork applied by customer service .



Installing

Install in the reverse order of removal, observing the following:



Note

Secure the loudspeaker using only the prescribed pop rivets ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

- Install pop rivets with pop rivet pliers - VAS 5072 A- .

2.6 Removing and installing loudspeaker for emergency call module - R335-

Removing

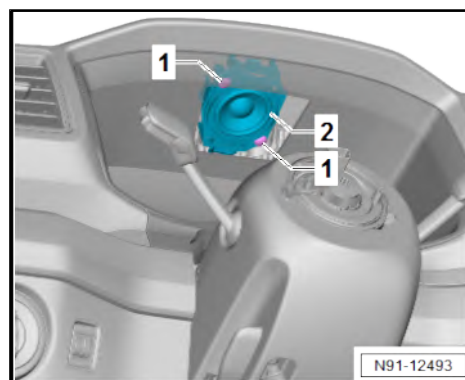
- Remove dash panel insert trim ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing dash panel insert trim .
- Remove loudspeaker trim ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing loudspeaker trim .
- Unscrew bolts -1- from loudspeaker -2-.
- Remove loudspeaker -2-.
- Separate electrical connector.

Installing

Install in the reverse order of removal, observing the following:

Specified torques

- ◆ Assembly overview - dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Assembly overview - dash panel



3 Aerial systems

⇒ ["3.1 Components of aerial systems", page 19](#)

⇒ ["3.2 Assembly overview - aerials", page 21](#)

⇒ ["3.3 Removing and installing roof aerial RX5 ", page 23](#)

⇒ ["3.4 Removing and installing aerial module", page 24](#)

3.1 Components of aerial systems

Components of aerial systems, for radio, radio navigation, Car-Net ⇒ [page 19](#)

Components of aerial systems, telematics preparation
⇒ [page 20](#)

Components of aerial systems, for radio, radio navigation, Car-Net

1 - Roof aerial - RX5-

- ☐ With following aerials depending on equipment:
- ☐ GPS aerial - R50- for Global Navigation Satellite System (GNSS)
- ☐ GSM aerial - R205- for telephone
- ☐ Radio aerial - R11- for AM/FM
- ☐ Digital radio aerial - R183-
- ☐ Removing and installing
⇒ [page 23](#)

2 - Wiring connection for AM/FM aerial

3 - Display unit for control device of front display and information control panel - J685-

- ☐ Composition Audio radio
- ☐ Removing and installing
⇒ [page 29](#)

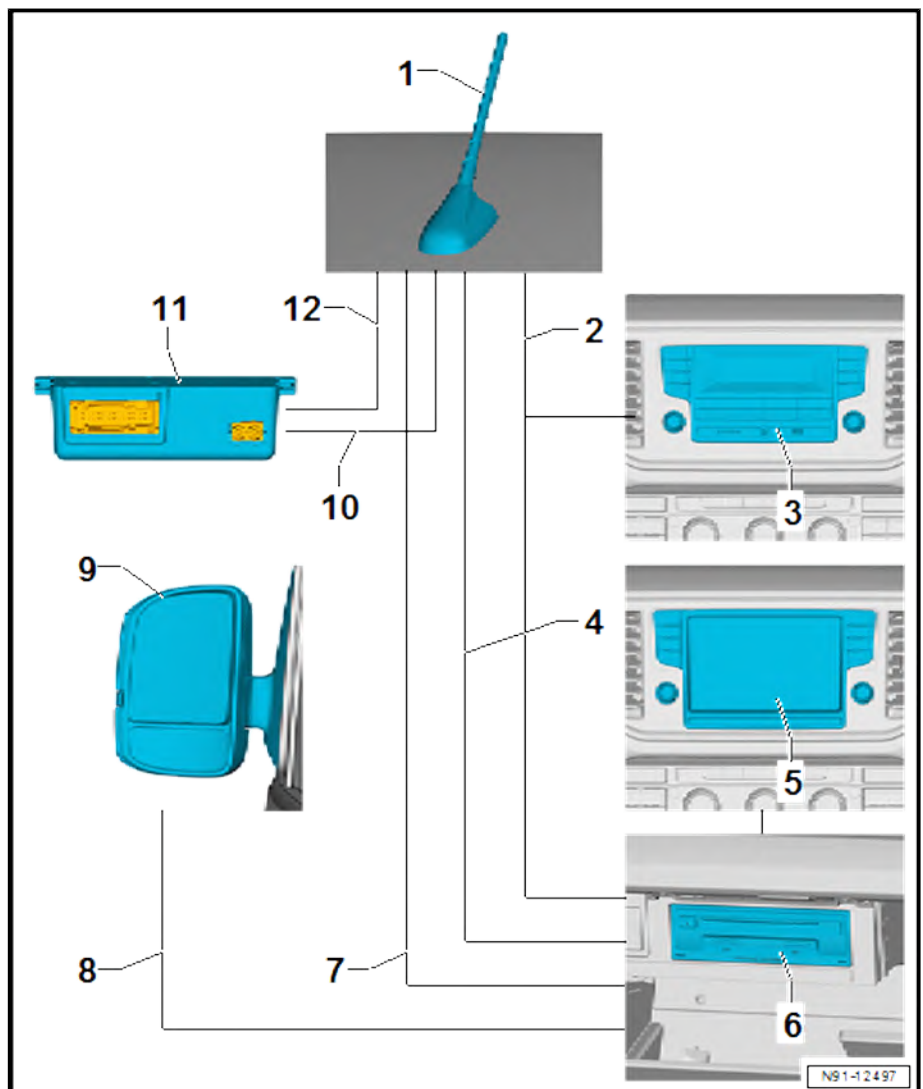
4 - Wiring connection for navigation aerial

5 - Display unit for control device of front display and information control panel - J685-

- ☐ Composition Media radio and Composition Media Discover radio
- ☐ Removing and installing
⇒ [page 29](#)

6 - Information electronics control unit 1 - J794-

- ☐ Composition Media radio and Composition Media Discover radio
- ☐ Removing and installing ⇒ [page 11](#)
- ☐ Pin assignment ⇒ Current flow diagrams, Electrical fault finding and Fitting locations



7 - Digital radio reception aerial wiring

8 - FM2/diversity aerial wiring

9 - Left exterior mirror

- ☐ With radio aerial - R248- for FM2
- ☐ With GSM aerial - R205- for telephone
- ☐ Removing and installing aerial module ⇒ [page 24](#)

10 - Wiring connection for navigation aerial for emergency call module control unit and communication unit - J949-

11 - Emergency call module control unit and communication unit - J949-

- ☐ With wiring connection for GSM aerial - R205- for telephone and GPS aerial - R50- to roof aerial - RX5-
- ☐ Removing and installing ⇒ [page 67](#)

12 - Wiring connection for GSM aerial - R205- for telephone to emergency call module control unit and communication unit - J949-

Components of aerial systems, telematics preparation

1 - Right exterior mirror

- ☐ with GPS aerial - R50- for navigation
- ☐ with GSM aerial - R205- for telephone
- ☐ Removing and installing aerial module ⇒ [page 24](#)

2 - Wiring connection for GPS aerial - R50- for navigation

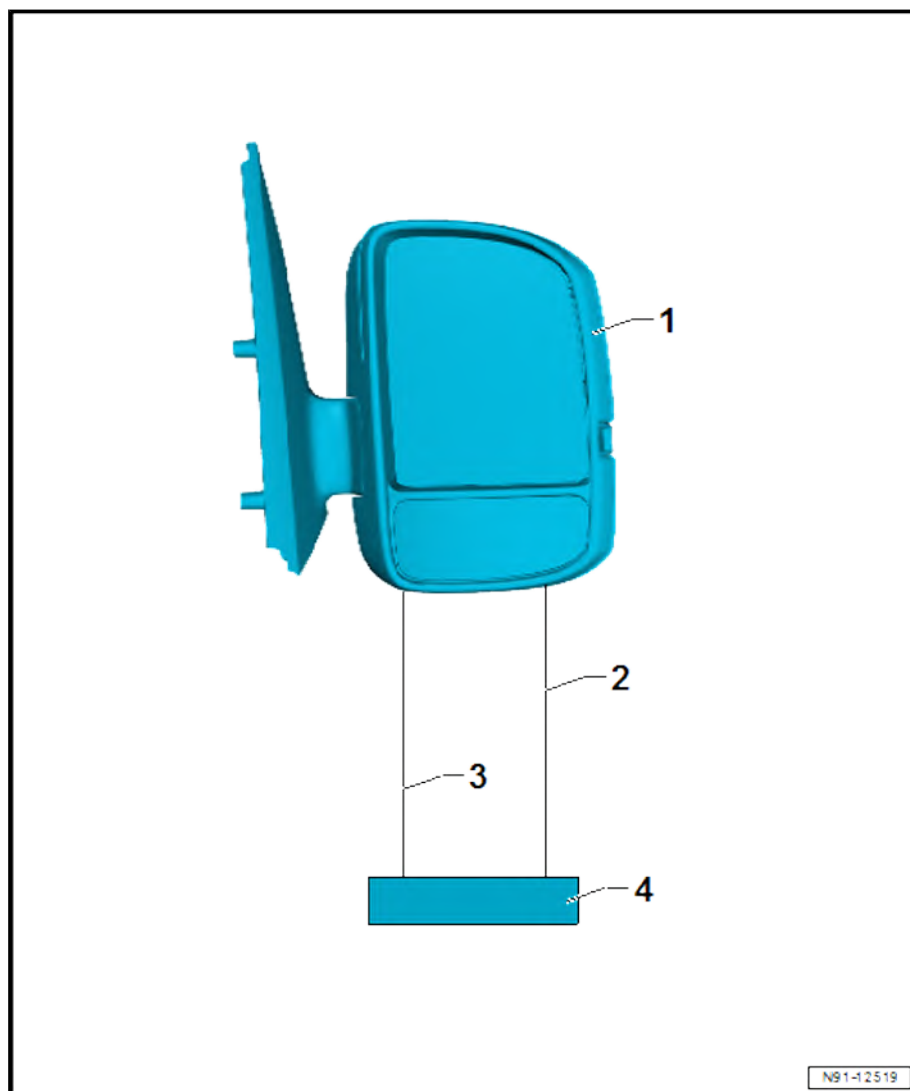
- ☐ to electrical connector for Fleet Management Interface (FSM) for telematics preparation
- ☐ In front passenger footwell behind dash panel cover

3 - Wiring connection GSM aerial - R205- for telephone

- ☐ to electrical connector for Fleet Management Interface (FSM) for telematics preparation
- ☐ In front passenger footwell behind dash panel cover

4 - Electrical connector for Fleet Management Interface (FSM) for telematics preparation

- ☐ In front passenger footwell behind dash panel cover





3.2 Assembly overview - aerials

Assembly overview - aerials, roof aerial ⇒ [page 21](#)

Assembly overview - aerials, left exterior mirror ⇒ [page 21](#)

Assembly overview - aerials, right exterior mirror ⇒ [page 22](#)

Assembly overview - aerials, roof aerial

1 - Roof aerial - RX5-

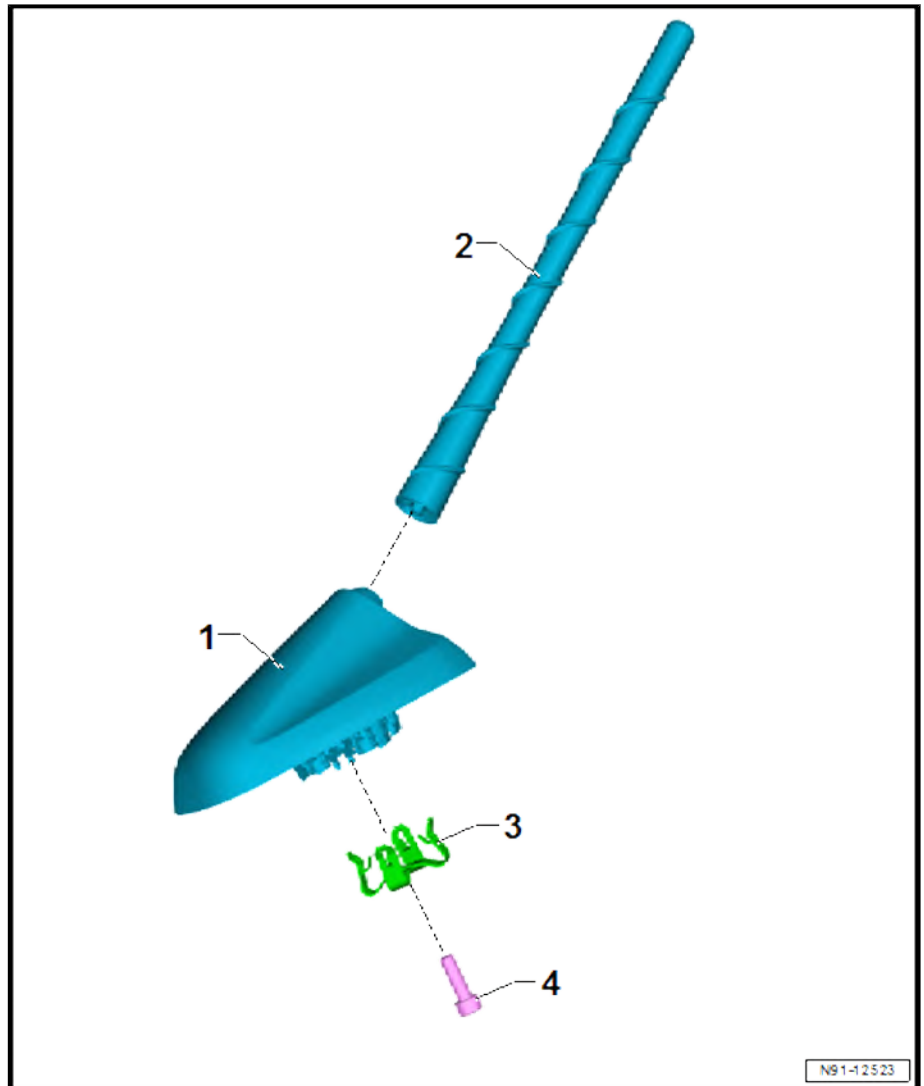
- ☐ Radio aerial - R11- for AM/FM
- ☐ Digital radio aerial - R183- for digital radio reception
- ☐ GPS aerial - R50- for Global Navigation Satellite System (GNSS)
- ☐ GSM aerial - R205- telephone aerial for emergency call system only
- ☐ Removing and installing ⇒ [page 23](#)

2 - Aerial rod

3 - Bracket

4 - Bolt

- ☐ 6 Nm



Assembly overview - aerials, left exterior mirror

1 - Mirror cover, top

- ❑ Removing and installing
⇒ General body repairs,
exterior; Rep. gr. 66 ;
Exterior mirror; Remov-
ing and installing mirror
cover .

2 - Radio aerial - R248- for FM2 (diversity aerial)

- ❑ Removing and installing
⇒ [page 24](#)

3 - GSM aerial - R205- tele- phone aerial

- ❑ Removing and installing
⇒ [page 24](#)

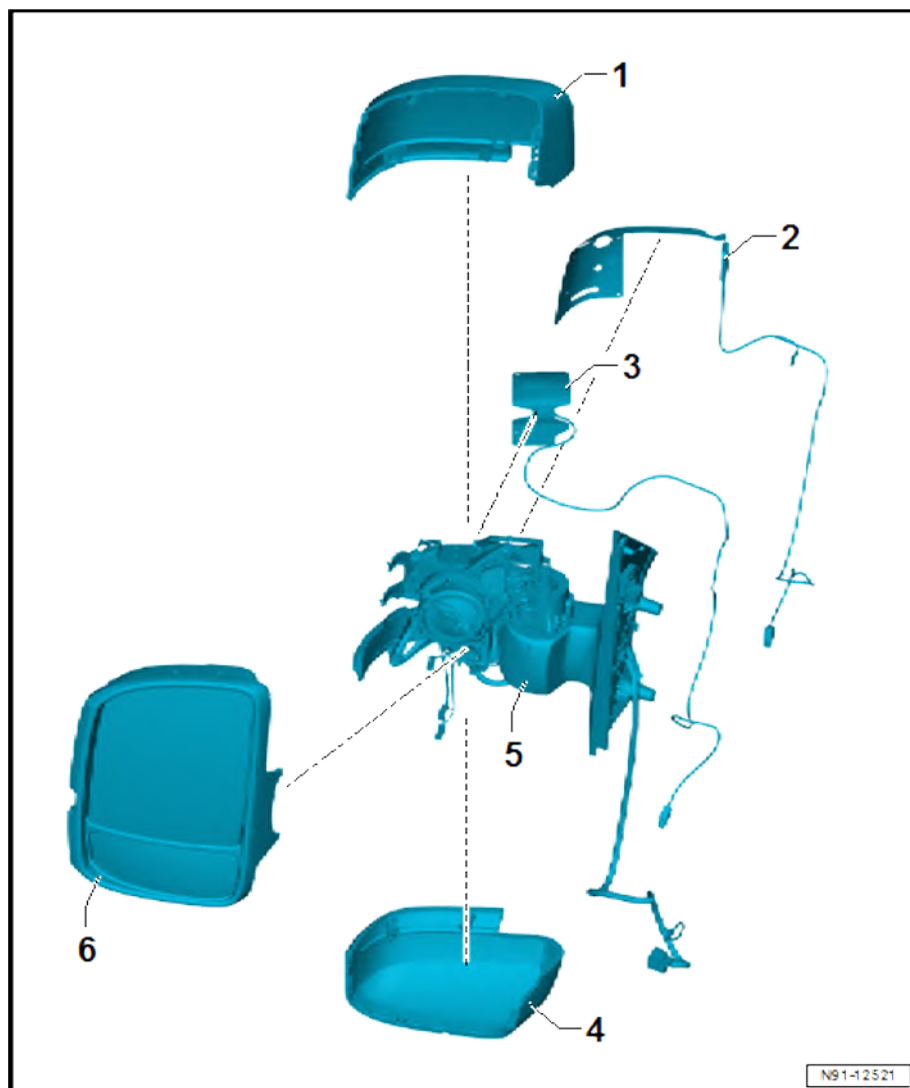
4 - Mirror cover, bottom

- ❑ Removing and installing
⇒ General body repairs,
exterior; Rep. gr. 66 ;
Exterior mirror; Remov-
ing and installing mirror
cover .

5 - Mirror base

- ❑ Removing and installing
⇒ General body repairs,
exterior; Rep. gr. 66 ;
Exterior mirror; Remov-
ing and installing exteri-
or mirror .

6 - Front frame with mirror glass



Assembly overview - aerials, right exterior mirror



1 - GPS aerial - R50-

- ☐ Aerial for Global Navigation Satellite System (GNSS)
- ☐ Only for telematics preparation, depending on equipment; for more information, refer to [⇒ page 61](#)
- ☐ Removing and installing [⇒ page 26](#)

2 - GSM aerial - R205-

- ☐ Telephone aerial
- ☐ Only for telematics preparation, depending on equipment; for more information, refer to [⇒ page 61](#)
- ☐ Removing and installing [⇒ page 26](#)

3 - Mirror cover, top

- ☐ Removing and installing [⇒ General body repairs, exterior; Rep. gr. 66 ; Exterior mirror; Removing and installing mirror cover .](#)

4 - Mirror base

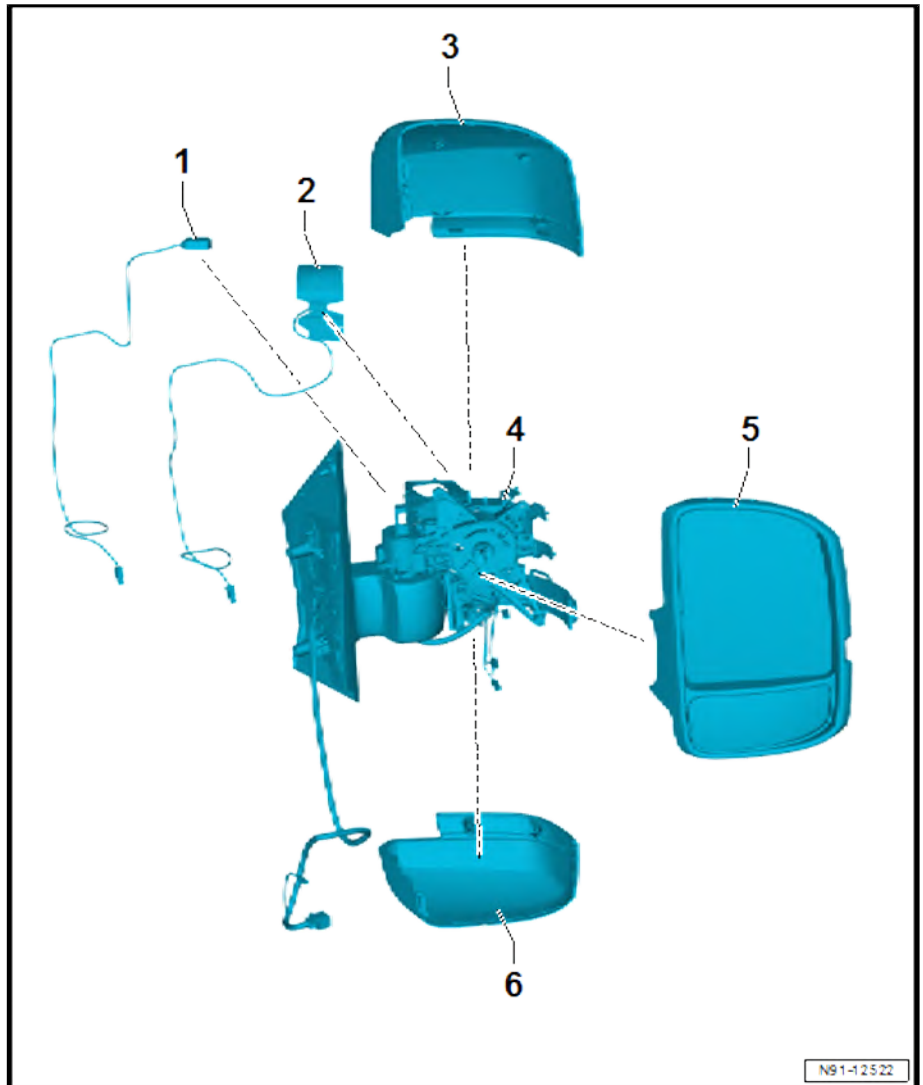
- ☐ Removing and installing [⇒ General body repairs, exterior; Rep. gr. 66 ; Exterior mirror; Removing and installing exterior mirror .](#)

5 - Front frame with mirror glass

- ☐ Removing and installing [⇒ General body repairs, exterior; Rep. gr. 66 ; Exterior mirror; Removing and installing mirror glass .](#)

6 - Mirror cover, bottom

- ☐ Removing and installing [⇒ General body repairs, exterior; Rep. gr. 66 ; Exterior mirror; Removing and installing mirror cover .](#)



3.3 Removing and installing roof aerial - RX5-

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

- Switch off ignition and all electrical consumers.

Vehicles without roof compartment

- Remove front moulded headliner ⇒ General body repairs, interior; Rep. gr. 70 ; Roof trim; Removing and installing moulded headliner .

Continued for all vehicles

- Disconnect connectors of aerial wires on aerial base.
- Unscrew bolt -3- on aerial base -1- and remove with retaining clip -2-.
- Remove roof aerial - RX5- .

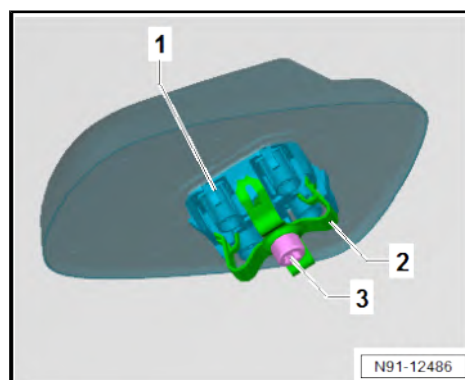
Installing

Install in the reverse order of removal, observing the following:

- Make sure that the rubber seal is seated correctly at the aerial base.
- Position and press on roof aerial - RX5- until it engages.

Specified torques

- ◆ ⇒ [“3.2 Assembly overview - aerals”, page 21](#)



3.4 Removing and installing aerial module

⇒ [“3.4.1 Removing and installing aerial module, radio aerial R248”, page 24](#)

⇒ [“3.4.2 Removing and installing aerial module GSM aerial R205”, page 25](#)

⇒ [“3.4.3 Removing and installing aerial module GPS aerial R50”, page 26](#)

⇒ [“3.4.4 Removing and installing aerial module GSM aerial R205”, page 27](#)

3.4.1 Removing and installing aerial module, radio aerial - R248-



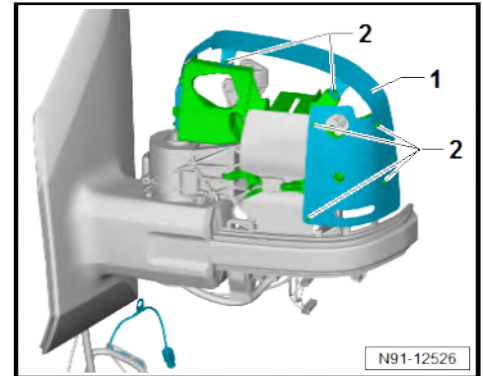
Note

Fitting location: in left exterior mirror (for LHD and RHD vehicles)



Removing

- Remove door trim ⇒ General body repairs, interior; Rep. gr. 70 ; Front door trims; Removing and installing front door trim .
- Remove exterior mirror ⇒ General body repairs, exterior; Rep. gr. 66 ; Exterior mirrors; Removing and installing exterior mirror .
- Remove mirror cover ⇒ General body repairs, exterior; Rep. gr. 66 ; Exterior mirrors; Removing and installing mirror cover .
- Unclip radio aerial - R248- -1- in mirror housing from securing elements -2-.

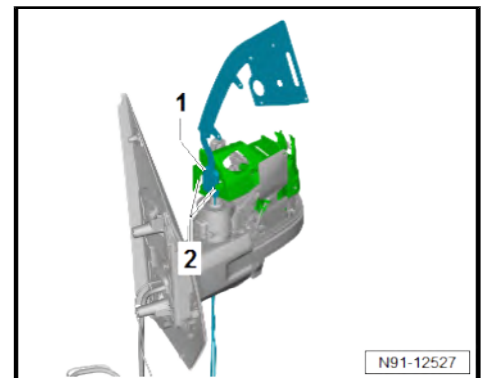


- Unclip radio aerial - R248- -1- from retainer -2-.
- Guide out aerial wiring connections from aerial housing.

Installing

Install in the reverse order of removal, observing the following:

- Route and secure wiring connections in mirror housing so that original condition is restored.



3.4.2 Removing and installing aerial module GSM aerial - R205-



Note

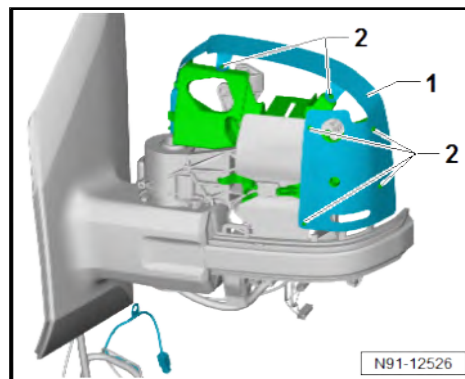
Fitting location: in left exterior mirror (for LHD and RHD vehicles)

Removing

- Remove door trim ⇒ General body repairs, interior; Rep. gr. 70 ; Front door trims; Removing and installing front door trim .
- Remove exterior mirror ⇒ General body repairs, exterior; Rep. gr. 66 ; Exterior mirrors; Removing and installing exterior mirror .
- Remove mirror cover ⇒ General body repairs, exterior; Rep. gr. 66 ; Exterior mirrors; Removing and installing mirror cover .



- Unclip radio aerial - R248- -1- in mirror housing from securing elements -2-.

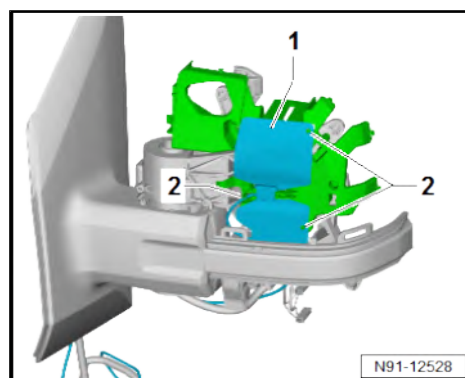


- Unclip GSM aerial - R205- -1- in mirror housing from securing elements -2-.
- Guide out aerial wiring connections from aerial housing.

Installing

Install in the reverse order of removal, observing the following:

- Route and secure wiring connections in mirror housing so that original condition is restored.



3.4.3 Removing and installing aerial module GPS aerial - R50-



Note

Fitting location: in left exterior mirror (for LHD and RHD vehicles)

Removing

- Remove door trim ⇒ General body repairs, interior; Rep. gr. 70 ; Front door trims; Removing and installing front door trim .
- Remove exterior mirror ⇒ General body repairs, exterior; Rep. gr. 66 ; Exterior mirrors; Removing and installing exterior mirror .
- Remove mirror cover ⇒ General body repairs, exterior; Rep. gr. 66 ; Exterior mirrors; Removing and installing mirror cover .



Note

The aerial is bonded to the mirror housing carrier.

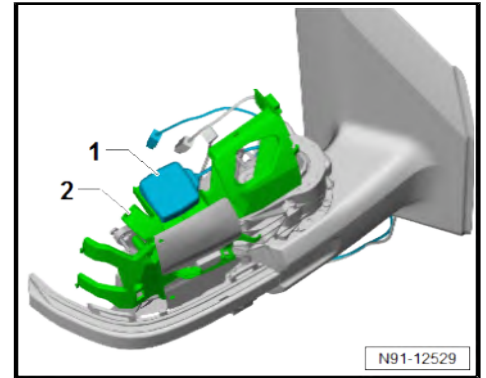


- Carefully detach GPS aerial - R50- -1- in mirror housing from mirror housing carrier -2- using a suitable screwdriver.
- Guide out aerial wiring connections from aerial housing.

Installing

Install in the reverse order of removal, observing the following:

- Pull off backing paper from adhesive pad and bond GPS aerial - R50- to mirror housing carrier.
- Route and secure wiring connections in mirror housing so that original condition is restored.



3.4.4 Removing and installing aerial module GSM aerial - R205-



Note

- ◆ *Fitting location: in right exterior mirror (for LHD and RHD vehicles)*
- ◆ *This specific GSM aerial - R205- is intended to be used only in the equipment version "Telematics".*

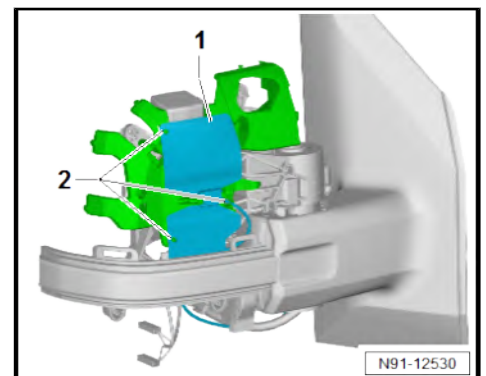
Removing

- Remove door trim ⇒ General body repairs, interior; Rep. gr. 70 ; Front door trims; Removing and installing front door trim .
- Remove exterior mirror ⇒ General body repairs, exterior; Rep. gr. 66 ; Exterior mirrors; Removing and installing exterior mirror .
- Remove mirror cover ⇒ General body repairs, exterior; Rep. gr. 66 ; Exterior mirrors; Removing and installing mirror cover .
- Unclip GSM aerial - R205- -1- in mirror housing from securing elements -2-.
- Guide out aerial wiring connections from aerial housing.

Installing

Install in the reverse order of removal, observing the following:

- Route and secure wiring connections in mirror housing so that original condition is restored.



4 Radio

⇒ ["4.1 Components of radio system", page 28](#)

⇒ ["4.2 Assembly overview - radio", page 28](#)

⇒ ["4.3 Removing and installing radio", page 29](#)

⇒ ["4.4 Pin assignments", page 30](#)

4.1 Components of radio system

For information on radios, refer to
⇒ ["1 Infotainment system", page 3](#) .

4.2 Assembly overview - radio

1 - Display unit for control device of front display and information control panel - J685-

- ❑ Removing and installing
⇒ [page 29](#)
- ❑ Connector pin assignment
⇒ [page 30](#) .

2 - Mounting frame for radio/navigation system

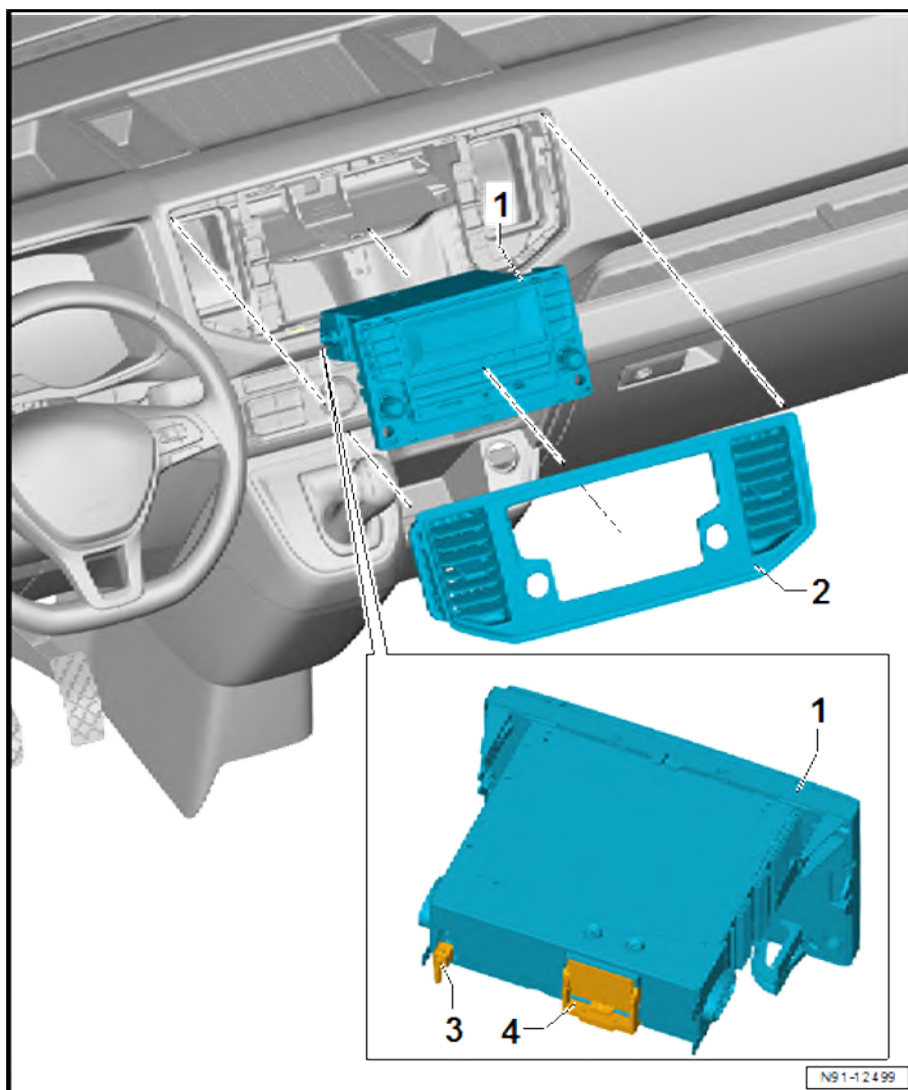
- ❑ Removing and installing
⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing dash panel vents

3 - Aerial connection

- ❑ Connector pin assignment
⇒ [page 30](#) .

4 - Multi-pin connector

- ❑ Connector pin assignment
⇒ [page 30](#) .





4.3 Removing and installing radio

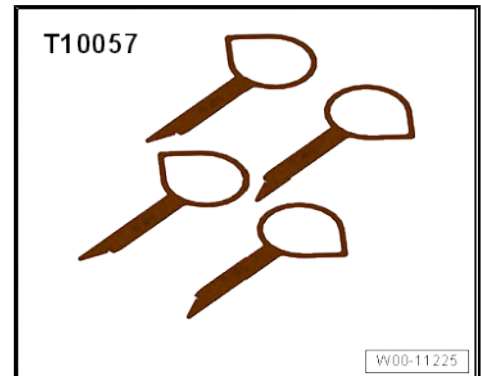
⇒ [“4.3.1 Removing and installing radio, Composition Audio version”, page 29](#)

⇒ [“4.3.2 Removing and installing radio, Composition Media and Discover Media”, page 30](#)

4.3.1 Removing and installing radio, Composition Audio version

Special tools and workshop equipment required

- ◆ Radio release tool - T10057-



Removing

- Switch off ignition and all electrical consumers.
- Remove centre vent ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing centre vent .
- Insert radio release tools - T10057- -1- into 4 release slots of radio.



Note

The pointed projections on the grip rings of radio release tools - T10057- must point outwards.

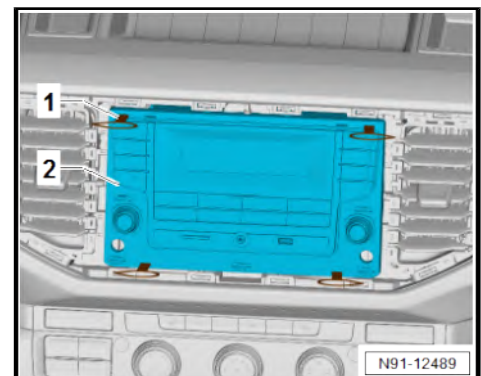
- Release upper locking elements by pressing them in direction of travel.
- Release lower locking elements by pressing them in direction of travel.
- Pull out radio -2-.
- Separate electrical connectors.

Installing

Install in the reverse order of removal, observing the following:

When installing, do not press on the display and the control elements.

Radio must engage audibly during installation.





4.3.2 Removing and installing radio, Composition Media and Discover Media

The radio is part of control unit 1 for information electronics - J794- .

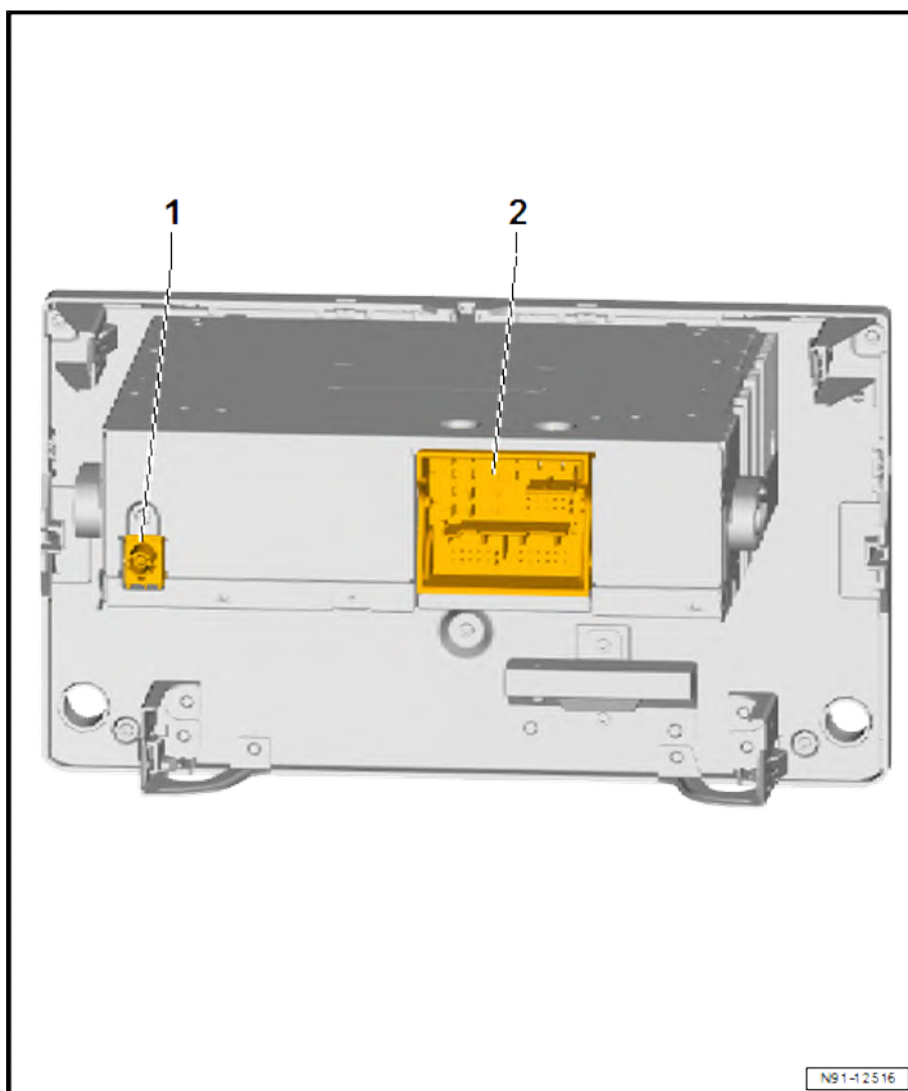
- Removing and installing control unit 1 for information electronics - J794- ➔ [page 11](#) .

4.4 Pin assignments

The pin assignment for the Composition Audio version is described in the following. Pin assignment for all other versions ➔ Current flow diagrams, Electrical fault finding and Fitting locations

1 - Connection for radio aerial
- R11-

2 - Connector block with 5 multi-pin connectors



Note

Contacts which are not listed are vacant/unassigned.



Connector block with 5 multi-pin connectors

A - connector, 8-pin - T8e-

- 1 - Loudspeaker (+), rear right
- 2 - Loudspeaker (+), front right
- 3 - Loudspeaker (+), front left
- 4 - Loudspeaker (+), rear left
- 5 - Loudspeaker (-), rear right
- 6 - Loudspeaker (-), front right
- 7 - Loudspeaker (-), front left
- 8 - Loudspeaker (-), rear left

B - connector, 12-pin - T12c-

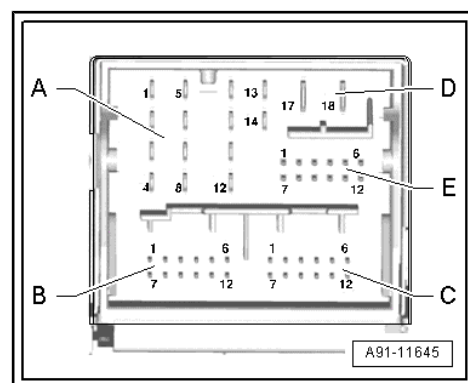
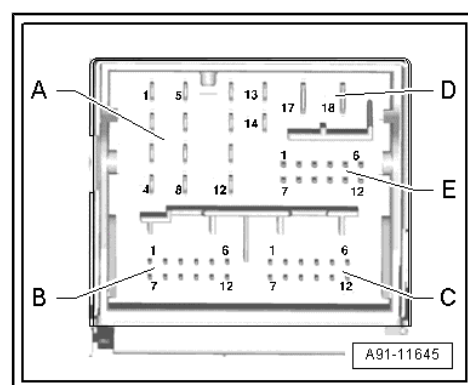
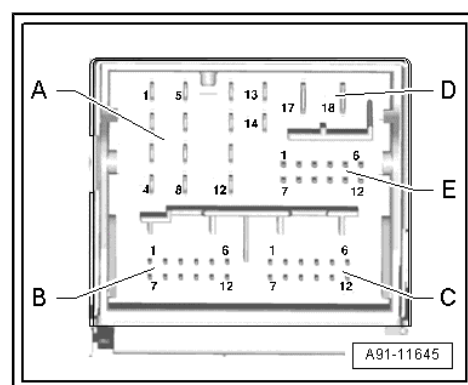
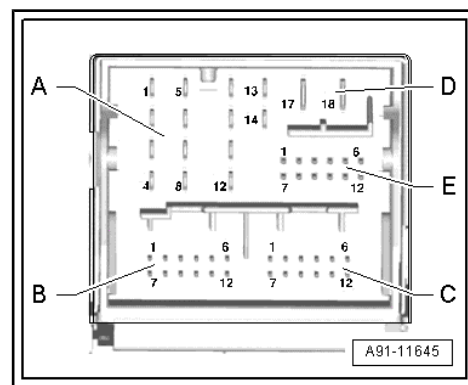
- 3 - Emergency call service mute

C - connector, 12-pin - T12e-

This connector is not used.

D - connector, 8-pin - T8f-

- 17 - Terminal 31
- 18 - Terminal 30



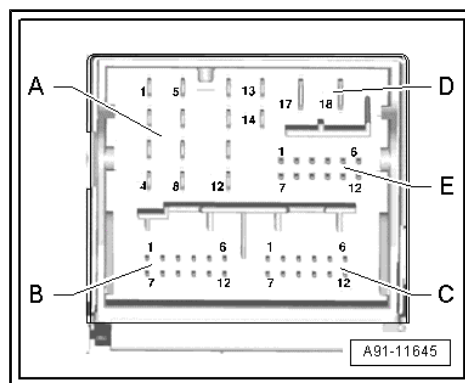


E - connector, 12-pin - T12d-

6 - Infotainment CAN bus, high

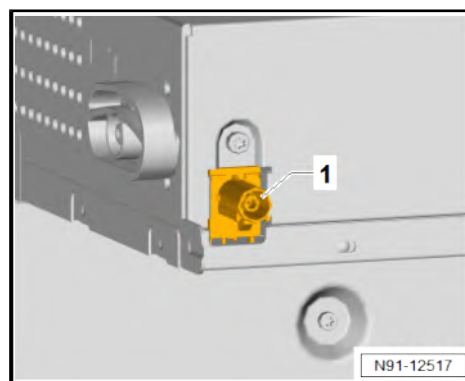
12 - Infotainment CAN bus, low

Aerial connection



AM/FM aerial connection

1 - Radio aerial - R11-





5 Telephone system

⇒ [“5.1 Mobile telephone preparation”, page 33](#)

⇒ [“5.2 Overview of fitting locations - telephone system”, page 34](#)

⇒ [“5.3 Removing and installing internal microphone R74 ”, page 35](#)

5.1 Mobile telephone preparation

⇒ [“5.1.1 Mobile telephone preparation, Bluetooth hands-free system”, page 33](#)

⇒ [“5.1.2 Mobile telephone preparation, Bluetooth car phone online”, page 33](#)

⇒ [“5.1.3 Notes on Bluetooth technology”, page 34](#)

5.1.1 Mobile telephone preparation, Bluetooth hands-free system

Bluetooth hands-free system is integrated in control unit 1 for information electronics - J794- on following radio/radio navigation systems:

- ◆ Composition Media
- ◆ Composition Media Discover

Bluetooth hands-free system is integrated in following radios:

- ◆ Composition Audio

Connection to mobile telephone - R54- is established via Bluetooth.

Bluetooth aerial - R152- is integrated in control unit 1 for information electronics - J794- or display unit for front information display and operating unit control unit - J685- depending on version.

Hands-free system consists of the following components:

- ◆ Display unit for front information display and operating unit control unit - J685-
- ◆ Control unit 1 for information electronics - J794- with aerial for Bluetooth - R152-
- ◆ Internal microphone - R74-

Repairing aerial wires ⇒ Electrical system; General information; Rep. gr. 97 ; Wiring harness and connector repair; Repairing aerial wires .



Note

Fault finding is performed using ⇒ Vehicle diagnostic tester.

5.1.2 Mobile telephone preparation, Bluetooth car phone online

Bluetooth car phone is integrated in control unit 1 for information electronics - J794- .

Connection to mobile telephone - R54- is established via Bluetooth.



Bluetooth aerial - R152- is integrated in control unit 1 for information electronics - J794- or display unit for front information display and operating unit control unit - J685- depending on version.

A WLAN hotspot is integrated to connect WLAN devices. A relevant mobile phone contract is a prerequisite for use.

Hands-free system consists of the following components:

- ◆ Display unit for front information display and operating unit control unit - J685-
- ◆ Control unit 1 for information electronics - J794- with aerial for Bluetooth - R152-
- ◆ Internal microphone - R74-

Repairing aerial wires ⇒ Electrical system; General information; Rep. gr. 97 ; Wiring harness and connector repair; Repairing aerial wires .



Note

Fault finding is performed using ⇒ Vehicle diagnostic tester.

5.1.3 Notes on Bluetooth technology

A standardised wireless connection is used for data transfer between control unit 1 for information electronics - J794- or display unit for front information display and operating unit control unit - J685- and mobile telephone - R54- - Bluetooth technology.

An additional sender/receiver unit is installed in control unit 1 for information electronics - J794- or display unit for front information display and operating unit control unit - J685- . Aerial for Bluetooth - R152- connects control unit 1 for information electronics - J794- to mobile telephone - R54- so that mobile telephone - R54- can be used without a wiring connection, i.e. wirelessly.

The transmission range is approx. 10 m.

5.2 Overview of fitting locations - telephone system



1 - Left exterior mirror with aerial system

- ☐ Removing and installing
⇒ [page 19](#)

2 - Multifunction buttons on left in steering wheel - E440-

- ☐ Removing and installing
⇒ [page 55](#)

3 - Multifunction buttons on right in steering wheel - E441-

- ☐ Removing and installing
⇒ [page 55](#)

4 - Display unit for control device of front display and information control panel - J685-

- ☐ As radio unit with Composition Audio
- ☐ As operating and display unit for multimedia system - R238- for Composition Media and Composition Media Discover
- ☐ Removing and installing
⇒ [page 29](#)

5 - Interior microphone - R74-

- ☐ In front interior light - WX1-
- ☐ Removing and installing
⇒ [page 35](#)

6 - Roof aerial - RX5-

- ☐ Removing and installing
⇒ [page 23](#)

7 - Information electronics control unit 1 - J794-

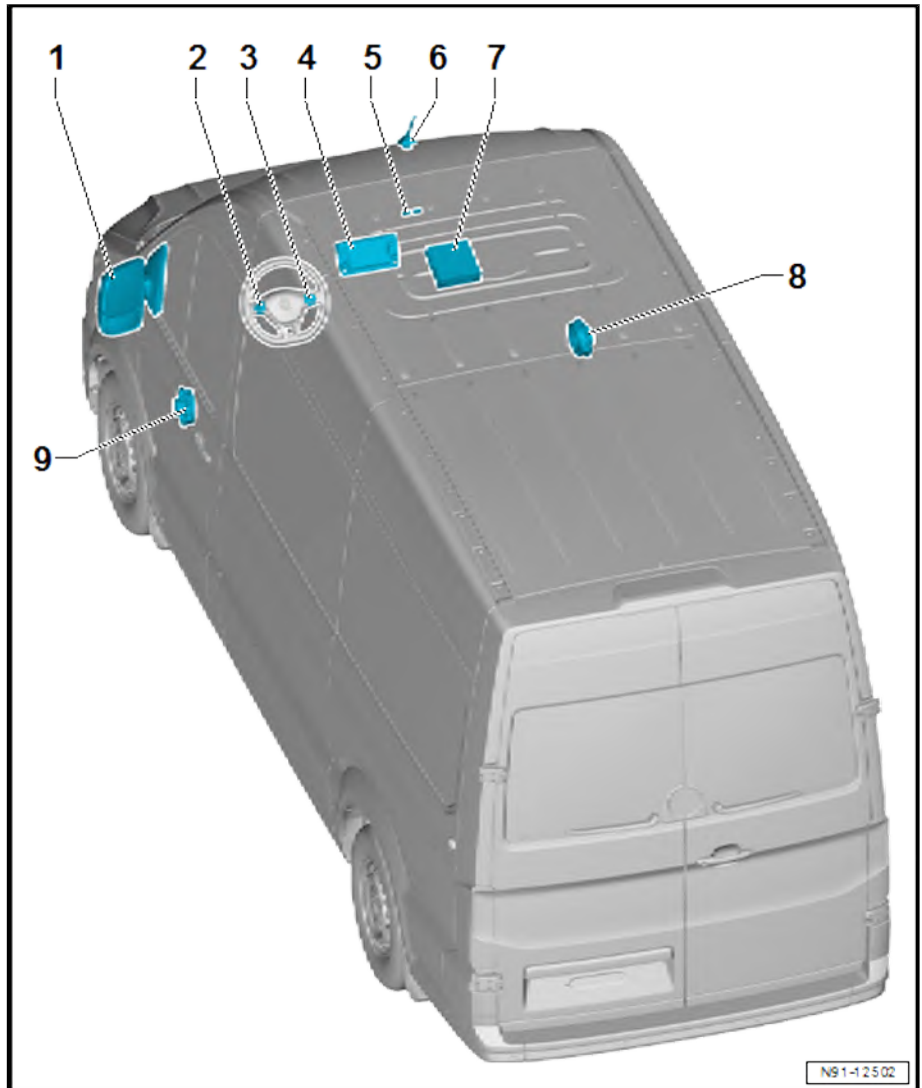
- ☐ In glove compartment.
- ☐ Only with Composition Media and Composition Media Discover
- ☐ Removing and installing ⇒ [page 11](#)
- ☐ Pin assignment ⇒ Current flow diagrams, Electrical fault finding and Fitting locations

8 - Sound system, right

- ☐ Removing and installing ⇒ [page 13](#)

9 - Sound system, left

- ☐ Removing and installing ⇒ [page 13](#)



5.3 Removing and installing internal microphone - R74-

Removing

- Switch off ignition and all electrical consumers.
- Remove front interior light - W1- ⇒ Electrical system; Rep. gr. 96 ; Lights; Removing and installing front interior light W1 .
- Remove storage compartment trim ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Removing and



installing headliner storage compartment; Removing and installing storage compartment trim .

- Release retaining hooks -3-.
- Remove internal microphone - R74- -2- from storage compartment in headliner -1-.

Installing

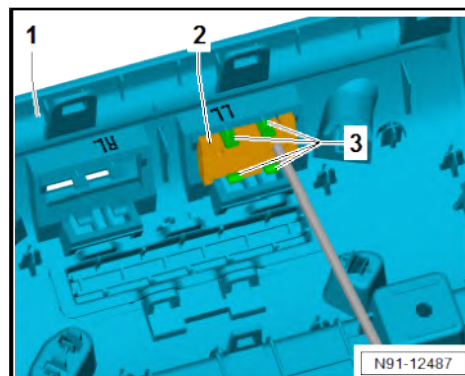
Install in the reverse order of removal, observing the following:

Internal microphone - R74- must engage audibly in storage compartment of headliner.



Note

Take note of the fitting locations for the internal microphone - R74- in the storage compartment of the headliner. The fitting locations are marked on the storage compartment with RL for right-hand and LL for left-hand drive vehicles.





6 Navigation system

⇒ [“6.1 Components of navigation system”, page 37](#)

⇒ [“6.2 Overview of fitting locations - navigation system”, page 37](#)

⇒ [“6.3 Removing and installing control unit for navigation system”, page 38](#)

6.1 Components of navigation system

The navigation function is integrated in control unit 1 for information electronics - J794- .

The navigation data are stored on a SD card.

Additional information ⇒ [page 3](#) .



Note

Fault finding is performed using ⇒ Vehicle diagnostic tester.

6.2 Overview of fitting locations - navigation system

1 - Left exterior mirror with aerial system

- ❑ Removing and installing
⇒ [page 19](#)

2 - Front left treble loudspeaker - R20-

- ❑ In front A-pillar
- ❑ Removing and installing
⇒ [page 15](#)

3 - Connection for external multimedia devices

- ❑ Removing and installing
⇒ [page 60](#)

4 - Multifunction buttons on left in steering wheel - E440-

- ❑ Removing and installing
⇒ [page 55](#)

5 - Multifunction buttons on right in steering wheel - E441-

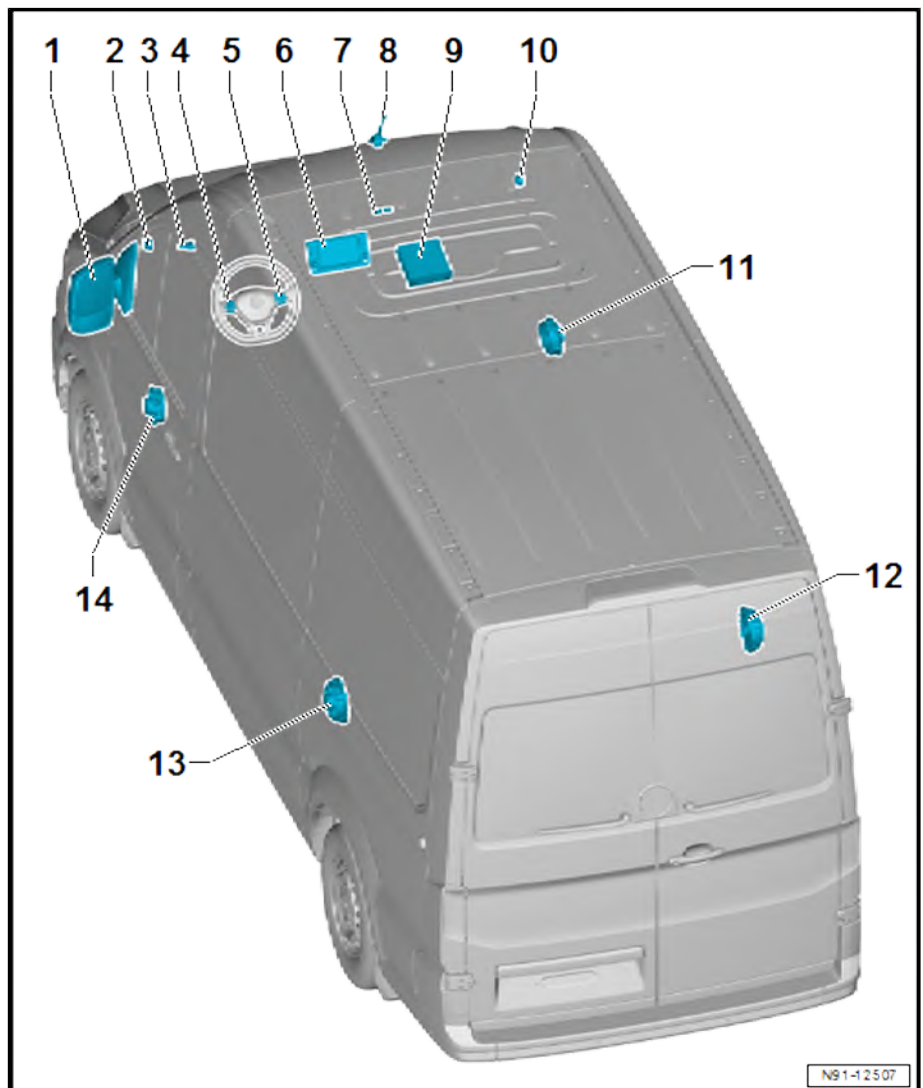
- ❑ Removing and installing
⇒ [page 55](#)

6 - Operating and display unit for multimedia system - R238-

- ❑ Removing and installing
⇒ [page 10](#)
- ❑ Pin assignment ⇒ Current flow diagrams, Electrical fault finding and Fitting locations

7 - Interior microphone - R74-

- ❑ Removing and installing
⇒ [page 35](#)





8 - Roof aerial - RX5-

- ☐ Removing and installing ⇒ [page 19](#)

9 - Information electronics control unit 1 - J794-

- ☐ Removing and installing ⇒ [page 11](#)
- ☐ Pin assignment ⇒ Current flow diagrams, Electrical fault finding and Fitting locations

10 - Front right treble loudspeaker - R22-

- ☐ In front A-pillar
- ☐ Removing and installing ⇒ [page 15](#)

11 - Bass loudspeaker, front right - R23-

- ☐ in front door trim
- ☐ Removing and installing ⇒ [page 17](#)

12 - Bass loudspeaker, rear right - R17-

- ☐ in side panel trim
- ☐ Removing and installing ⇒ [page 15](#)

13 - Bass loudspeaker, rear left - R15-

- ☐ in side panel trim
- ☐ Removing and installing ⇒ [page 15](#)

14 - Bass loudspeaker, front left - R21-

- ☐ in front door trim
- ☐ Removing and installing ⇒ [page 17](#)

6.3 Removing and installing control unit for navigation system

The control unit for navigation system is integrated in control unit 1 for information electronics - J794- .

- Removing and installing control unit 1 for information electronics - J794- ⇒ [page 11](#) .



7 Reversing camera system

⇒ [“7.1 Components of reversing camera system”, page 39](#)

⇒ [“7.2 Overview of fitting locations - reversing camera system”, page 43](#)

⇒ [“7.3 Assembly overview - reversing camera”, page 45](#)

⇒ [“7.4 Removing and installing reversing camera R189”, page 46](#)

⇒ [“7.5 Removing and installing Trailer Assist camera R368”, page 47](#)

⇒ [“7.6 Calibrating Trailer Assist camera R368”, page 47](#)

7.1 Components of reversing camera system

⇒ [“7.1.1 Components of reversing camera for parking aid”, page 39](#)

⇒ [“7.1.2 Components of reversing camera for Trailer Assist”, page 40](#)

7.1.1 Components of reversing camera for parking aid

The reversing camera supports the driver during reversing by providing the driver with an image of the traffic situation behind the vehicle via display unit for control device of front display and information control panel - J685- .

The system is switched on by engaging reverse gear.

The reversing camera system consists of the following components:

- ◆ Reversing camera - R189-
- ◆ Information electronics control unit 1 - J794-
- ◆ Display unit for front information display and operating unit control unit - J685-



Note

- ◆ *When dealing with complaints, it is absolutely necessary to understand the function and operation of the reversing camera.*
- ◆ *Additional information can be found in the ⇒ Operating manual .*
- ◆ *Calibration of the reversing camera is not necessary after repairs to the vehicle.*
- ◆ *The reversing camera does not have a diagnosis function.*

Overview of reversing camera system for parking aid

1 - Onboard supply control unit - J519-

- ❑ Removing and installing
⇒ Electrical system;
Rep. gr. 97 ; Control
units; Removing and in-
stalling onboard supply
control unit - J519- .

2 - Reversing light switch - F4-

3 - Reversing camera - R189-

Installed in high-level brake
light.

- ❑ Removing and installing
⇒ [page 46](#)

4 - Wiring connection for convenience CAN bus

5 - Wiring connection for infotainment CAN bus

6 - Information electronics control unit 1 - J794-

- ❑ Removing and installing
⇒ [page 11](#)
- ❑ Pin assignment ⇒ Current
flow diagrams,
Electrical fault finding
and Fitting locations

7 - Wiring connection for MIB CAN

8 - Display unit for control device of front display and information control panel - J685-

- ❑ Removing and installing
⇒ [page 10](#)

9 - Wiring connection for LVDS

10 - Automatic gearbox control unit - J217-

- ❑ Removing and installing ⇒ Rep. gr. 39 ; Gearbox control system, Removing and installing gearbox control unit

11 - ABS control unit - J104-

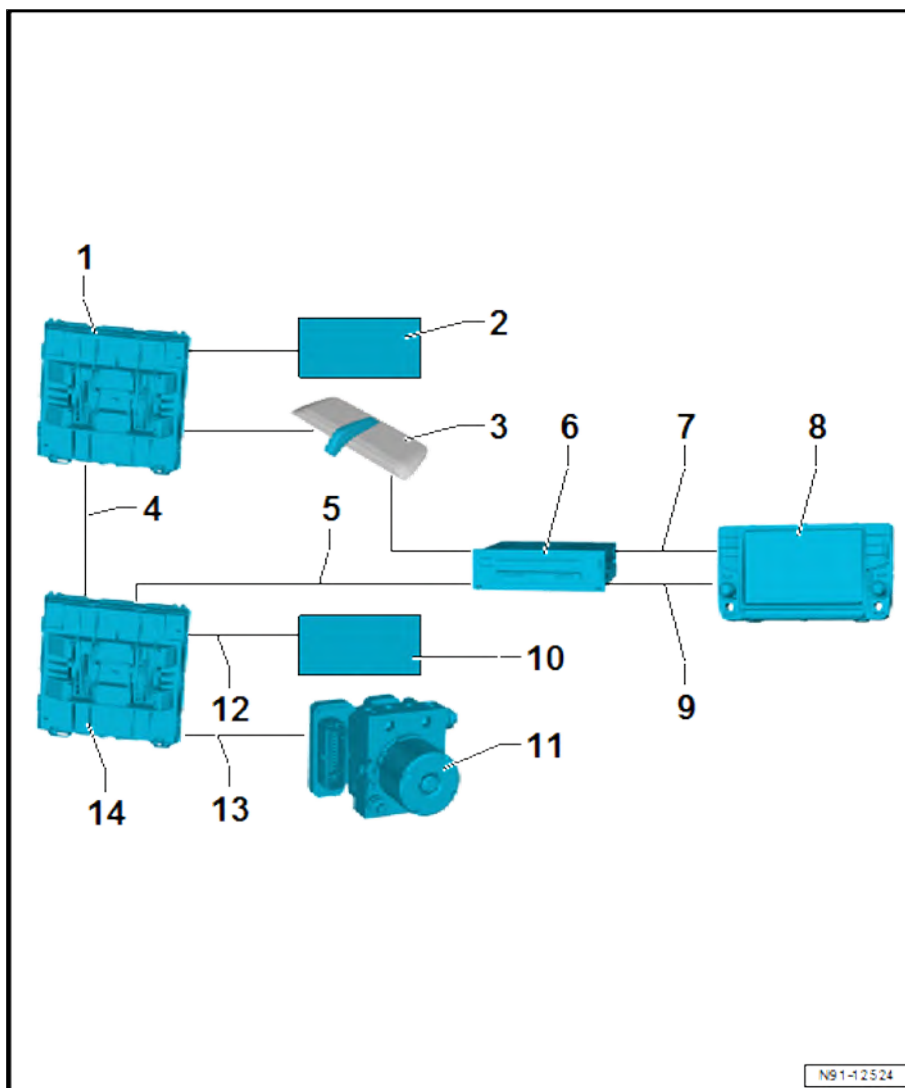
- ❑ Removing and installing ⇒ Brake system; Rep. gr. 45 ; Control unit and hydraulic unit; Removing and installing ABS control unit - J104- / ABS hydraulic unit - N55-

12 - Wiring connection for powertrain CAN bus

13 - Wiring connection for running gear CAN

14 - Diagnostics interface for data bus - J533-

- ❑ on fuse holder C - SC- on left beneath dash panel
- ❑ Removing and installing ⇒ Electrical system; Rep. gr. 97 ; Control units; Removing and installing data bus diagnostic interface - J533-



7.1.2 Components of reversing camera for Trailer Assist

Trailer Assist supports the driver when manoeuvring with the trailer. To do this, the driver must stop the vehicle, leave the engine running, engage reverse gear and activate the system using the park assist steering button - E581- or display unit for front infor-

mation display and operating unit control unit - J685- depending on the equipment level. The trailer must be connected to the vehicle electrically.

The Trailer Assist camera - R368- detects the current articulation angle of the trailer and indicates it on the display in the dash panel insert - KX2- . With the aid of the exterior mirror adjuster - EX11- , the driver can set the target direction of trailer travel by specifying the required articulation angle.

The system then takes control of the steering and performs the manoeuvre automatically. The driver just has to press the accelerator and brakes. The direction of the trailer can be corrected and readjusted during the entire driving manoeuvre.

For additional information refer to the ⇒ Owner's Manual of the vehicle.

Overview of reversing camera system for Trailer Assist

1 - Diagnostics interface for data bus - J533-

- ❑ Removing and installing
⇒ Electrical system;
Rep. gr. 97 ; Control
units; Removing and installing data bus diagnostic interface - J533-

2 - Wiring connection for convenience CAN bus

3 - Onboard supply control unit - J519-

- ❑ Removing and installing
⇒ Electrical system;
Rep. gr. 97 ; Control
units; Removing and installing onboard supply control unit - J519- .

4 - Driver door control unit - J386-

- ❑ Removing and installing
⇒ General body repairs, exterior; Rep. gr. 57 ; Central locking; Removing and installing driver door control unit - J386-

5 - Exterior mirror adjuster - EX11-

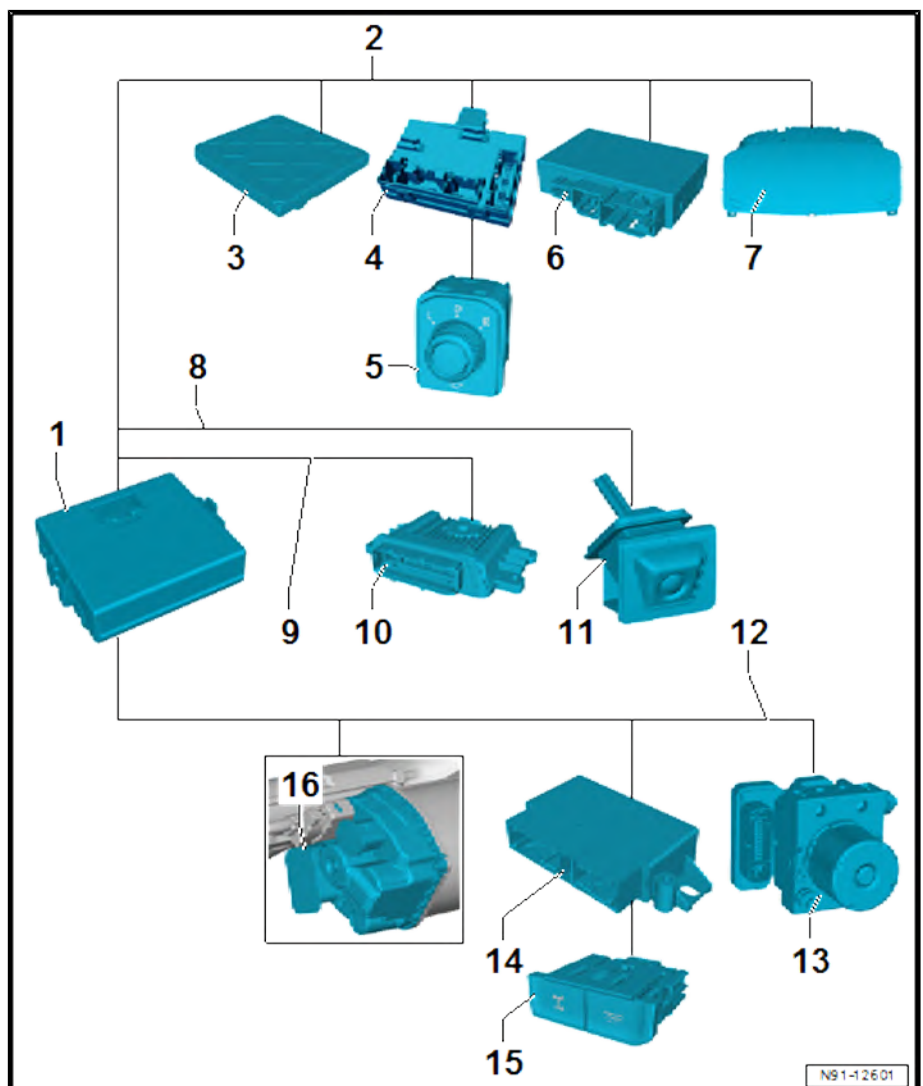
- ❑ With mirror adjustment switch - E43-
- ❑ Removing and installing
⇒ Electrical system;
Rep. gr. 96 ; Controls;
Removing and installing exterior mirror adjuster - EX11-

6 - Trailer detector control unit - J345-

- ❑ Removing and installing ⇒ Electrical system; Rep. gr. 94 ; Tow hitch; Removing and installing trailer detector control unit - J345-

7 - Multifunction indicator - J119-

- ❑ In dash panel insert - KX2-





- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 90 ; Dash panel insert; Removing and installing dash panel insert - KX2-

8 - Wiring connection for infotainment CAN bus

9 - Wiring connection for powertrain CAN bus

10 - Automatic gearbox control unit - J217-

- ☐ Removing and installing ⇒ Rep. gr. 39 ; Gearbox control system, Removing and installing gearbox control unit

11 - Trailer Assist camera - R368-

- ☐ Calibrating camera for trailer manoeuvring system - R368- ⇒ [page 47](#)
- ☐ Removing and installing ⇒ [page 47](#)

12 - Wiring connection for running gear CAN bus

13 - ABS control unit - J104-

- ☐ Removing and installing ⇒ Brake system; Rep. gr. 45 ; Control unit and hydraulic unit; Removing and installing ABS control unit - J104- / ABS hydraulic unit - N55-

14 - Parking aid control unit - J446-

- ☐ Integrated in park assist steering control unit - J791-
- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 94 ; Parking aid; Removing and installing parking aid control unit - J446-

15 - Park assist steering button - E581-

- ☐ Integrated in switch module 2, next to dash panel insert - EX47-
- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 96 ; Controls; Removing and installing park assist steering button - E581-

16 - Power steering control unit - J500-

- ☐ Integrated in steering rack
- ☐ Removing and installing ⇒ Running gear, axles, steering; Rep. gr. 48 ; Steering rack; Removing and installing steering rack



7.2 Overview of fitting locations - reversing camera system

Overview of fitting locations - reversing camera system for parking aid

1 - Display unit for control device of front display and information control panel - J685-

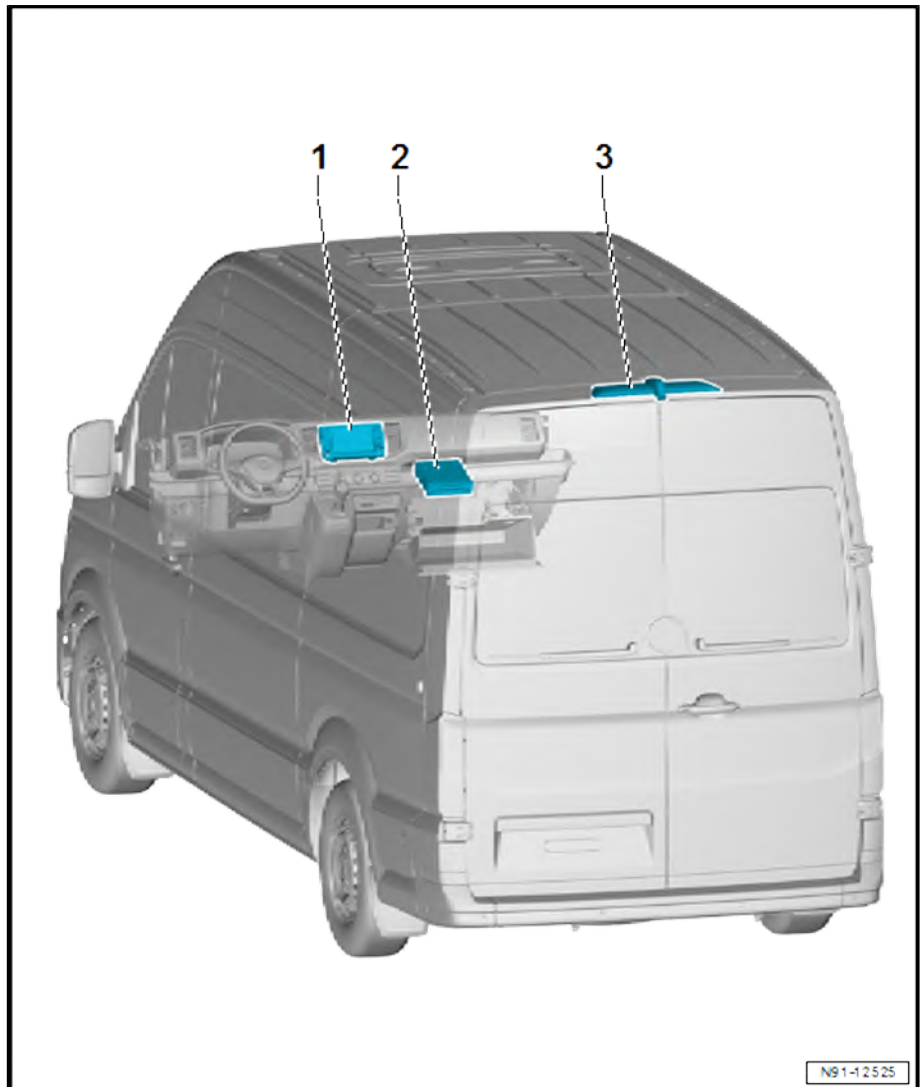
- ☐ In dash panel in the middle
- ☐ Removing and installing
⇒ [page 10](#)

2 - Information electronics control unit 1 - J794-

- ☐ In glove compartment.
- ☐ Removing and installing
⇒ [page 11](#)
- ☐ Pin assignment ⇒ Current flow diagrams, Electrical fault finding and Fitting locations

3 - Reversing camera - R189-

- ☐ Removing and installing
⇒ [page 46](#)



Overview of fitting locations - reversing camera system for Trailer Assist

1 - Mirror adjustment switch - E43-

- ☐ in exterior mirror adjuster - EX11- of driver door
- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 96 ; Controls; Removing and installing exterior mirror adjuster - EX11-

2 - Onboard supply control unit - J519-

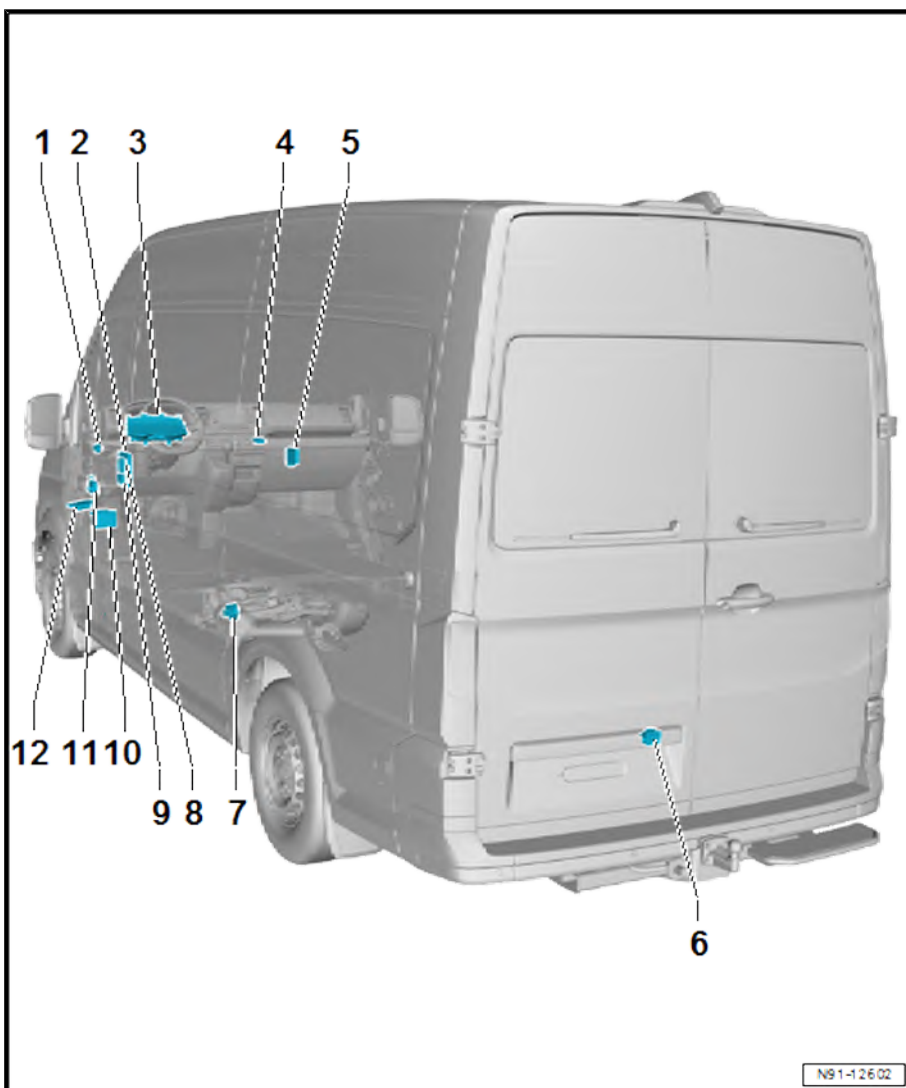
- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 97 ; Control units; Removing and installing onboard supply control unit - J519- .

3 - Multifunction indicator - J119-

- ☐ In dash panel insert - KX2-
- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 90 ; Dash panel insert; Removing and installing dash panel insert - KX2-

4 - Park assist steering button - E581-

- ☐ In switch module 2, next to dash panel insert - EX47-
- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 96 ; Controls; Removing and installing park assist steering button - E581-



5 - Trailer detector control unit - J345-

- ☐ Under dash panel on right
- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 94 ; Tow hitch; Removing and installing trailer detector control unit - J345-

6 - Trailer Assist camera - R368-

- ☐ In light strip in left wing door
- ☐ Calibrating camera for trailer manoeuvring system - R368- ⇒ [page 47](#)
- ☐ Removing and installing ⇒ [page 47](#)

7 - Power steering control unit - J500-

- ☐ Integrated in steering rack
- ☐ Removing and installing ⇒ Running gear, axles, steering; Rep. gr. 48 ; Steering rack; Removing and installing steering rack

8 - Diagnostics interface for data bus - J533-

- ☐ on fuse holder C - SC- on left beneath dash panel
- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 97 ; Control units; Removing and installing data bus diagnostic interface - J533-

9 - Parking aid control unit - J446-

- ☐ Integrated in park assist steering control unit - J791-
- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 94 ; Parking aid; Removing and installing parking aid control unit - J446-



10 - Driver door control unit - J386-

- ☐ In driver door
- ☐ Removing and installing ⇒ General body repairs, exterior; Rep. gr. 57 ; Central locking; Removing and installing driver door control unit - J386-

11 - ABS control unit - J104-

- ☐ Removing and installing ⇒ Brake system; Rep. gr. 45 ; Control unit and hydraulic unit; Removing and installing ABS control unit - J104- / ABS hydraulic unit - N55-

12 - Automatic gearbox control unit - J217-

- ☐ Removing and installing ⇒ Rep. gr. 39 ; Gearbox control system, Removing and installing gearbox control unit

7.3 Assembly overview - reversing camera

Assembly overview - reversing camera system for parking aid

1 - Camera housing

- ☐ Upper part

2 - Reversing camera - R189-

- ☐ Removing and installing
⇒ [page 46](#)

3 - Seal

4 - Seal

5 - Camera housing

- ☐ Lower part

6 - Bolts

- ☐ Qty. 2
- ☐ 1.5 Nm

7 - Cover plate

- ☐ For high-level brake light

8 - Seal

9 - Housing

- ☐ For high-level brake light
- ☐ Removing and installing
⇒ Electrical system;
Rep. gr. 94 ; High-level
brake light; Removing
and installing high-level
brake light

10 - Clips

- ☐ Qty. 8

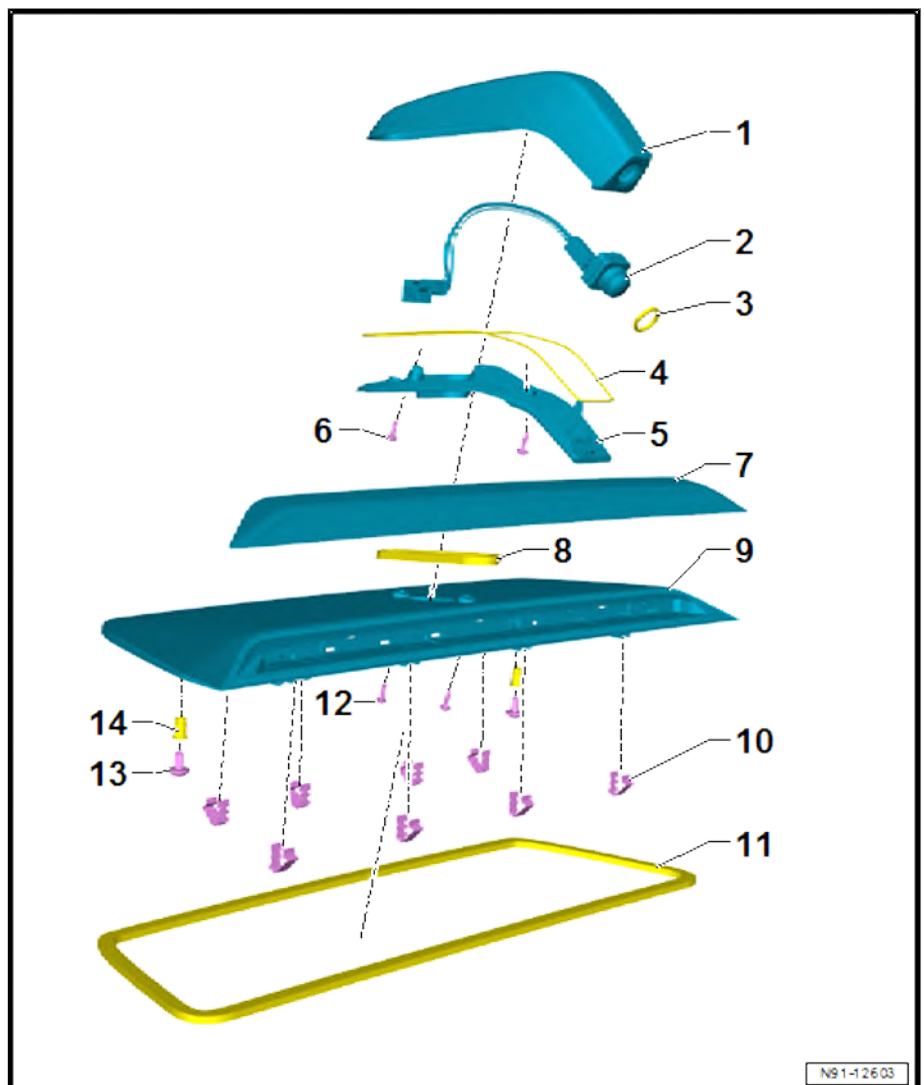
11 - Seal

12 - Bolts

- ☐ Qty. 2
- ☐ 1.5 Nm

13 - Bolts

- ☐ Qty. 2
- ☐ 1.5 Nm





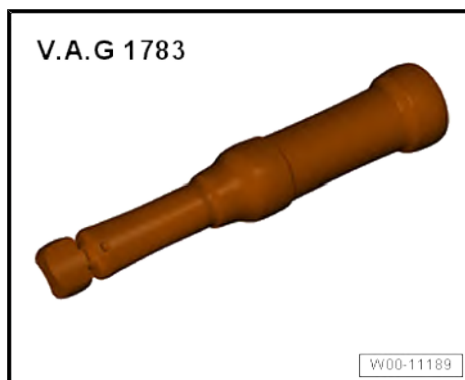
14 - Sleeves

- Qty. 2

7.4 Removing and installing reversing camera - R189-

Special tools and workshop equipment required

- ◆ Universal working platform
- ◆ Torque wrench - V.A.G 1783-



Removing

- Remove high-level brake light ⇒ Electrical system; Rep. gr. 94 ; High-level brake light; Removing and installing high-level brake light .
- Unscrew bolts -2-.
- Disconnect electrical connectors -3-.
- Remove reversing camera - R189- -1-.

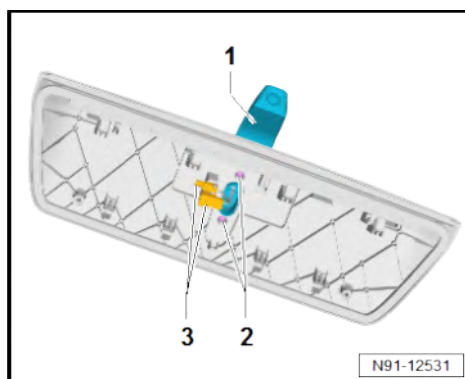
Installing

Install in the reverse order of removal, observing the following:



Note

There is no requirement to calibrate the reversing camera - R189- .



Specified torques

- ◆ ⇒ [“7.3 Assembly overview - reversing camera”, page 45](#)



7.5 Removing and installing Trailer Assist camera - R368-

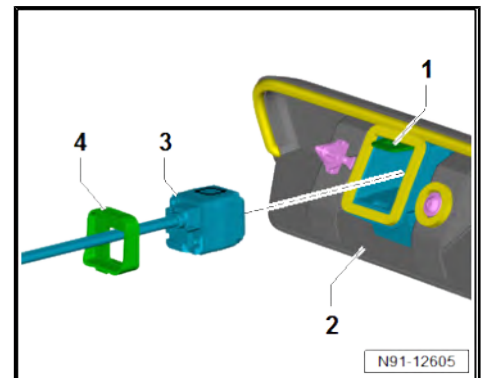
Special tools and workshop equipment required

- ◆ Removal wedge - 3409-



Removing

- Remove light strip ⇒ General body repairs, exterior; Rep. gr. 58 ; Wing door; Removing and installing door attachments .
- Raise tab -1- using removal wedge - 3409- and remove frame -4- from light strip -2-.
- Pull Trailer Assist camera - R368- -3- out of light strip -2-.



Installing

Install in the reverse order of removal, observing the following:



Note

- ◆ *The adhesive label on the camera housing must point upwards when installing.*
- ◆ *When the camera for trailer manoeuvring system - R368- is renewed or new software has been flashed, calibration of the camera for trailer manoeuvring system - R368- is required ⇒ [page 47](#) .*

7.6 Calibrating Trailer Assist camera - R368-

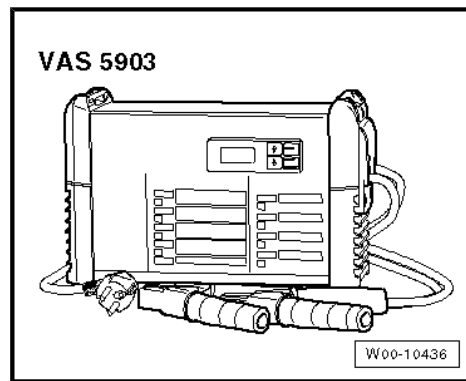
Special tools and workshop equipment required

- ◆ Calibration unit - VAS 6350A-





- ◆ Battery charger - VAS 5903-



- ◆ Vehicle diagnostic tester

The calibration unit - VAS 6350A- consists of the following components:

- ◆ Wheel centre mounting - VAS 6350A/1-
- ◆ Spacing laser - VAS 6350A/2-
- ◆ Linear laser - VAS 6350A/3-
- ◆ Tinted/coloured glasses - VAS 6350A/3A-

Notes on calibration

The reversing camera system for Trailer Assist may have to be recalibrated following repair work on the vehicle. More specifically, this is necessary after:

- ◆ Removing and installing reversing camera
- ◆ Removing and installing front axle
- ◆ Removing and installing rear axle
- ◆ Removing and installing light strip
- ◆ After accident repairs on wing door
- ◆ If new software was flashed
- ◆ After wheel alignment with adjustments
- ◆ Raising or lowering running gear
- ◆ Adjustment to overall length of vehicle e.g. by body builder

Calibration requirements

- ◆ Camera lens must be clean
- ◆ Vehicle must be positioned on a firm and level surface
- ◆ There must be sufficient space all around the vehicle
- ◆ Parking brake applied
- ◆ Steering wheel in zero position, wheels pointing straight ahead
- ◆ All doors and rear lid closed
- ◆ Nobody in vehicle
- ◆ Vehicle unladen (kerb weight)
- ◆ Battery charger - VAS 5903- connected

Overview of assembled calibration unit - VAS 6350A-



1 - Wheel centre mounting - VAS 6350A/1-

- ☐ Left and right

2 - Bracket

- ☐ Right and left
- ☐ Spacing laser - VAS 6350A/2- mounting

3 - Plastic foot

- ☐ Total of 3 on bottom of measuring equipment
- ☐ Adjustable to adjust horizontal position of measuring facility

4 - Linear laser - VAS 6350A/3-

- ☐ Switching on and off ⇒ Owner's manual

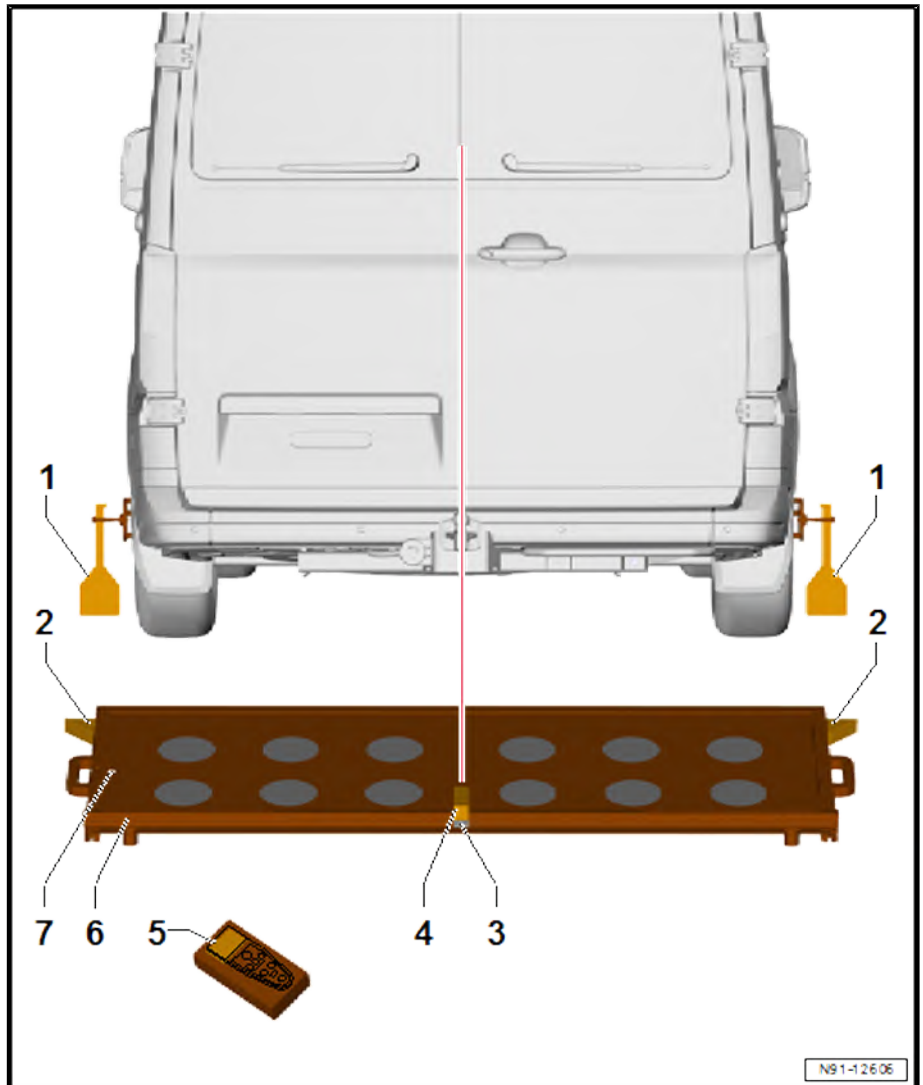
5 - Spacing laser - VAS 6350A/2-

- ☐ Notes on operation ⇒ Owner's manual

6 - Calibration unit - VAS 6350A-

7 - Spirit level

- ☐ On calibration board
- ☐ For checking horizontal position



Aligning the calibration board

- Position calibration unit - VAS 6350A- behind vehicle between angle brackets of calibration unit and paddles on wheel centre mountings - VAS 6350 A/1- of rear wheels, as shown in overview diagram.
- ◆ Distance between vehicle and calibration unit for all vehicles: 180 cm +/- 2 cm
- ◆ Distance between vehicle and calibration unit for vehicles with long overhang: 240 cm +/- 2 cm



- Turn plastic feet -3- under calibration unit - VAS 6350A- until air bubble is exactly in centre of indicator -arrow- (i.e. is completely level).



Note

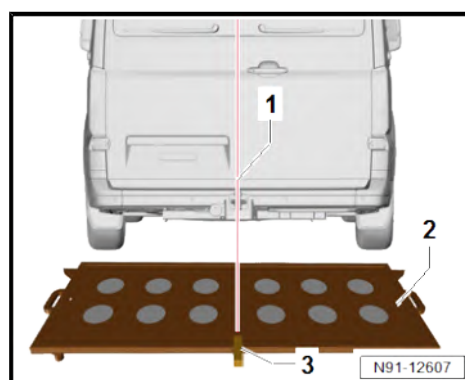
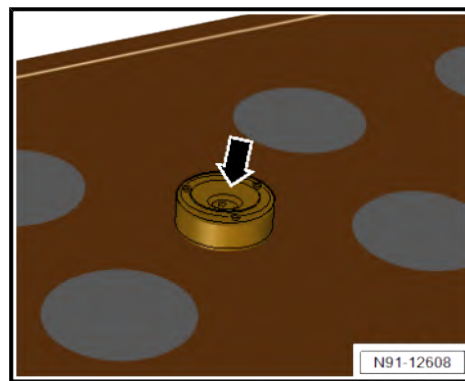
- ◆ *Make sure there is no light reflection on the calibration unit - VAS 6350A- .*
- ◆ *Light reflections will impair the image recognition of the Trailer Assist camera - R368- and might make calibration impossible.*
- ◆ *Do not look directly into the laser beam of the linear laser - VAS 6350A/3- . Risk of eye damage!*

- Switch on linear laser - VAS 6350A/3- -3-.
- Align calibration unit - VAS 6350A- -2- in such a way that laser beam -1- is focused on rear of vehicle in centre above manufacturer emblem.



Note

If owing to unfavourable lighting conditions the laser beam cannot be seen on the vehicle body, use tinted/coloured glasses - VAS 6350A/3A- .



Mount wheel centre mounting - VAS 6350A/1-

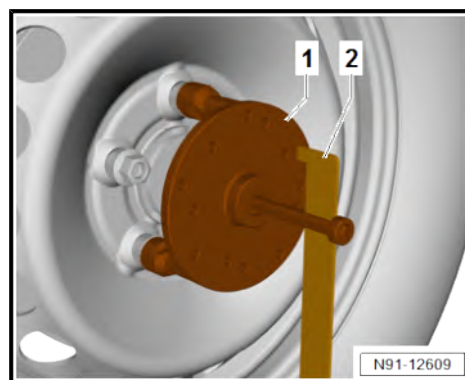
- Prepare wheel centre mountings - VAS 6350A/1- accordingly.
- Screw 3 × adapters for wheel securing bolts on hole pattern designated 120 at both wheel centre mountings - VAS 6350A/1- .
- Position paddles on both wheel centre mountings - VAS 6350A/1- , and secure each paddle with clamping bolt.
- Fit wheel centre mounting - VAS 6350A/1- -1- onto wheel securing bolts on rear axle.



Note

When fitting the wheel centre mountings - VAS 6350A/1- on the wheels, make sure that none of anti-theft wheel bolts (if applicable) is attached to the wheel centre mountings - VAS 6350A/1- .

- Adjust paddles -2- using clamping bolts so that they can move easily just above the floor.





Measuring dimensions



Note

Do not look directly into the laser beam of the spacing laser - VAS 6350A/2- . Risk of eye damage!

- Switch on spacing laser - VAS 6350A/2- .

Display -1- is shown.

The display -2- shows how the spacing laser - VAS 6350A/2- must be applied. Press the respective button.

1 - Fit with front edge -arrow-

3 - Fit with rear edge -arrow-

- Hold spacing laser - VAS 6350A/2- for distance measurement, as shown in illustration, on right of calibration unit - VAS 6350A- flush in bracket -2-. The spacing laser must be firmly in contact with the bracket.

- Press measurement button.

The laser is switched on.

Laser beam from spacing laser - VAS 6350A/2- must hit lower, larger part of paddle -1-.

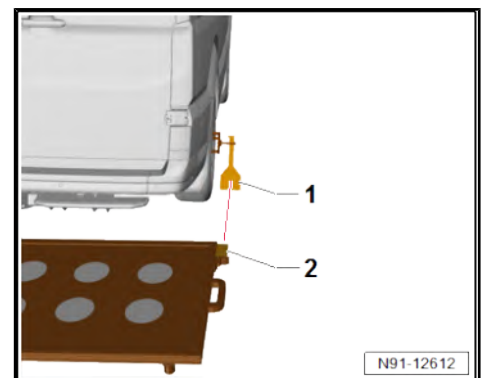
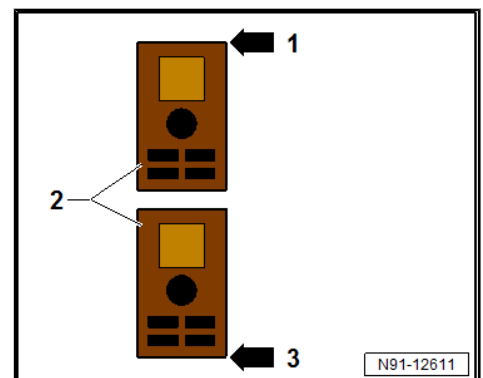
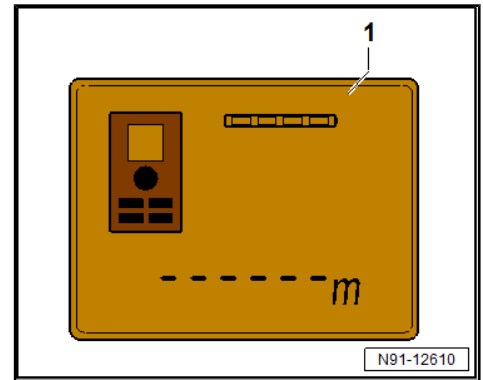
If this is not the case, position of paddles must be corrected accordingly using clamping bolts on wheel centre mounting - VAS 6350A/1- .

- Hold spacing laser - VAS 6350A/2- in locating bracket on calibration unit - VAS 6350A- with one hand while laser beam is visible on paddle.
- Press measuring button briefly to measure distance.
- Make a note of the displayed value.
- Perform same measuring procedure for left wheel on other side of calibration unit - VAS 6350A- .

Measured distance value must be identical on both sides.

- If value is not identical, adjust calibration unit - VAS 6350A- until values on both sides are identical.
- When adjusting, make sure that linear laser - VAS 6350A/3- hits centre of manufacturer emblem and spirit level display also remains centred.
- Adjust if necessary.

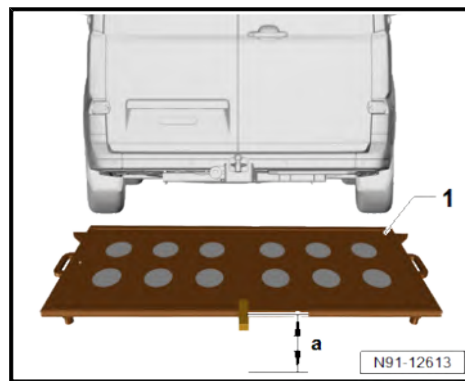
Value for measured distance must be entered in ⇒ Vehicle diagnostic tester during calibration.





- Measure height of calibration unit - VAS 6350A- -1-, dimension -a- (from upper edge of calibration unit - VAS 6350A- to wheel footprint).

Make sure that spacing laser - VAS 6350A/2- is set correctly (fitted with front edge).



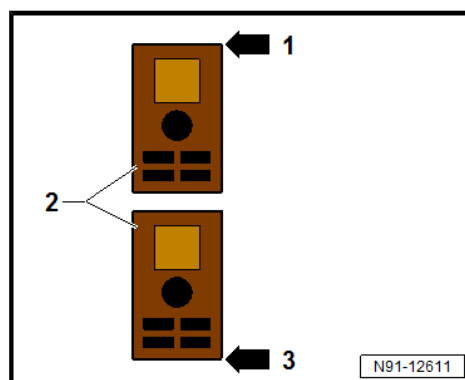
The display -2- shows how the spacing laser - VAS 6350A/2- must be applied. Press the respective button.

1 - Fit with front edge -arrow-

3 - Fit with rear edge -arrow-

- Note the reading for the height of the calibration unit - VAS 6350A- .

Value for measured distance must be entered in ⇒ Vehicle diagnostic tester during calibration.



Note

- ◆ *When determining the height of the camera, make sure that no hard objects get in contact with the camera lens to prevent it from being scratched.*
- ◆ *Since direct measurement on the vehicle is prevented by constructional restraints, the dimension is measured in two steps, with the two being added to make the total result.*



- Measure height of camera lens, dimension -a- and -b- (from centre of camera lens for Trailer Assist camera - R368- to wheel footprint).
- Use the spacing laser - VAS 6350A/2- or a commercially available folding rule to measure the distance from the centre of the camera lens to the upper edge of the bumper, dimension -a-.
- Measure distance from upper edge of bumper to wheel footprint, dimension -b-.
- Add both values and make a note of total result.

The measured distance, along with the height of the calibration unit - VAS 6350A- and the centre height of the Trailer Assist camera - R368- lens must be entered into the ⇒ Vehicle diagnostic tester in millimetres during calibration.

- Connect battery charger - VAS 5903- ⇒ Electrical system; Rep. gr. 27 ; Battery; Charging battery .



Note

If a software update or flashing or calibrations are performed on a vehicle, a charging unit with a charging current of at least 70A (e. g. battery charger - VAS 5903-) must be used to avoid problems with maintaining the power supply during the work.

Calibrating Trailer Assist camera - R368-

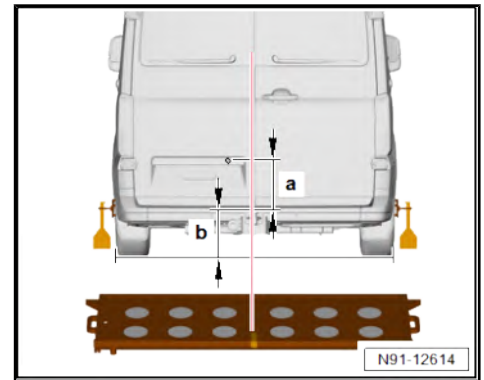


Note

First note the points regarding preliminary work for calibration ⇒ [page 47](#) .

- Connect ⇒ Vehicle diagnostic tester ⇒ Maintenance ; Booklet 10.3 ; General; Vehicle diagnostic tester .
- Switch on ignition.

Perform calibration using ⇒ Vehicle diagnostic tester.





8 Multifunction steering wheel

⇒ ["8.1 Components of multifunction steering wheel", page 54](#)

⇒ ["8.2 Assembly overview - multifunction steering wheel", page 54](#)

⇒ ["8.3 Removing and installing steering wheel multifunction buttons E441 / E440 ", page 55](#)

⇒ ["8.4 Removing and installing Tiptronic switch in steering wheel E439 / E438 ", page 58](#)

8.1 Components of multifunction steering wheel

For a better usability of infotainment, telephone, navigation, cruise control system and vehicle menu, buttons are integrated in steering wheel. For vehicles with Tiptronic, additional rocker switches are provided on right and left sides.

The control unit for multifunction steering wheel - J453- (multifunction buttons on left in steering wheel - E440-) reads the button information and transfers them to control unit for steering column electronics - J527- via LIN data bus. From control unit for steering column electronics - J527- the data are transferred to the individual devices via CAN bus (comfort) and data bus diagnostic interface - J533- .

Fault finding is performed using ⇒ Vehicle diagnostic tester.

8.2 Assembly overview - multifunction steering wheel



1 - Tiptronic switch in steering wheel to shift down - E439-

- ☐ Depending on equipment
- ☐ Removing and installing
⇒ [page 58](#)

2 - Steering wheel

- ☐ Removing and installing
⇒ Running gear, axles, steering; Rep. gr. 48 ; Steering wheel; Removing and installing steering wheel

3 - Tiptronic switch in steering wheel to shift up - E438-

- ☐ Depending on equipment
- ☐ Removing and installing
⇒ [page 58](#)

4 - Bolts

- ☐ Qty. 2
- ☐ 1 Nm

5 - Trim panel

- ☐ Removing and installing
⇒ [page 55](#)

6 - Multifunction buttons on right in steering wheel - E441-

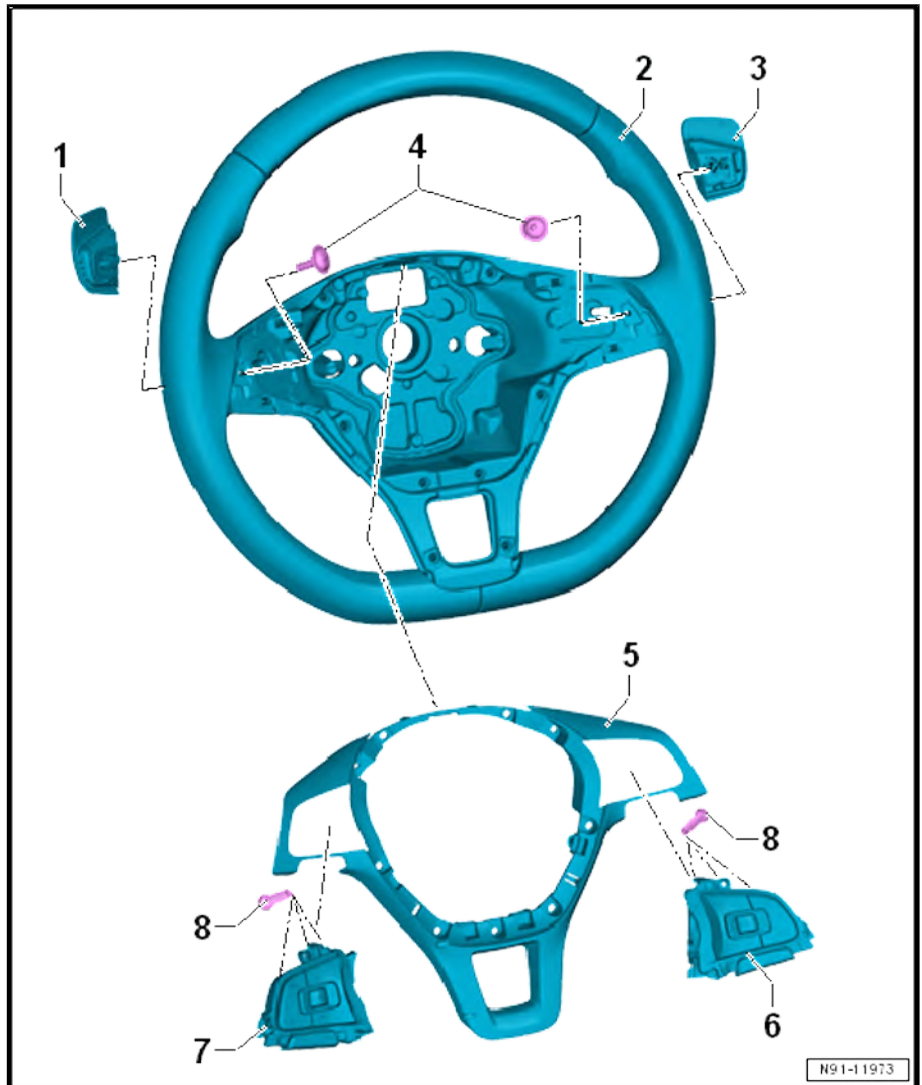
- ☐ Removing and installing
⇒ [page 55](#)

7 - Multifunction buttons on left in steering wheel - E440-

- ☐ Removing and installing
⇒ [page 55](#)

8 - Bolts

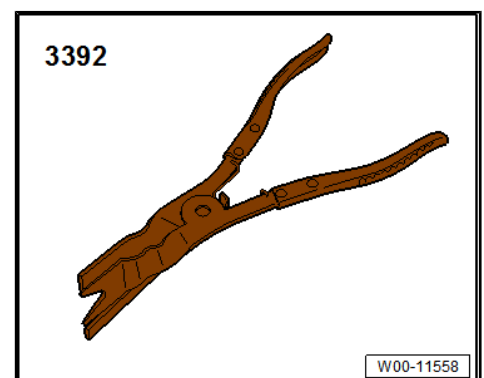
- ☐ Qty. 6
- ☐ 1 Nm



8.3 Removing and installing steering wheel multifunction buttons -E441- / -E440-

Special tools and workshop equipment required

- ◆ Removal pliers - 3392-





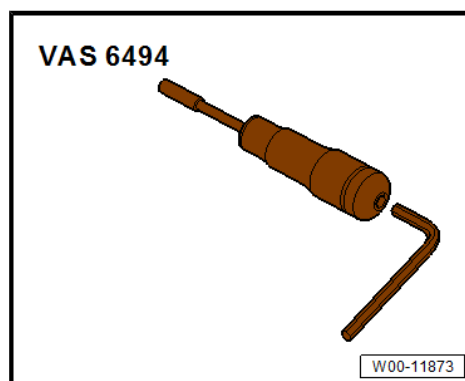
- ◆ Removal wedge - 3409- , qty. 2



- ◆ Lever - T10468-



- ◆ Torque screwdriver - VAS 6494-



Removing

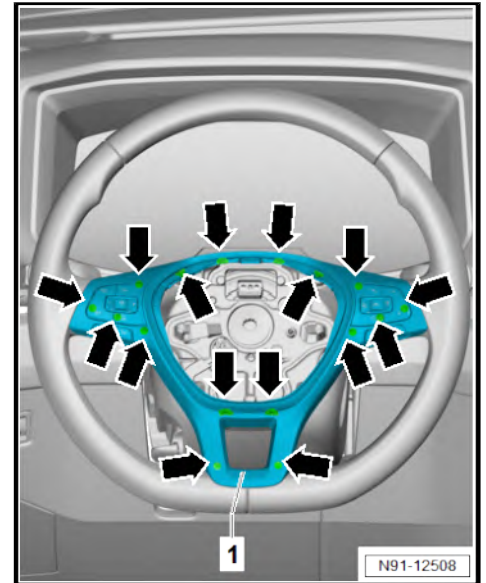


Note

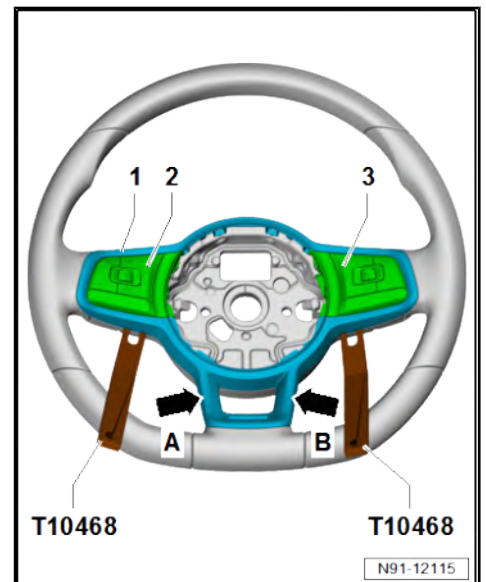
The multifunction buttons in the steering wheel -E441- / -E440- are screwed to the trim. The trim together with the multifunction buttons is engaged on the steering wheel. It is easier to remove the trim using two removal wedges - 3409- .

- Remove driver side airbag ⇒ General body repairs, interior; Rep. gr. 69 ; Driver side airbag; Removing and installing air-bag unit with igniter .

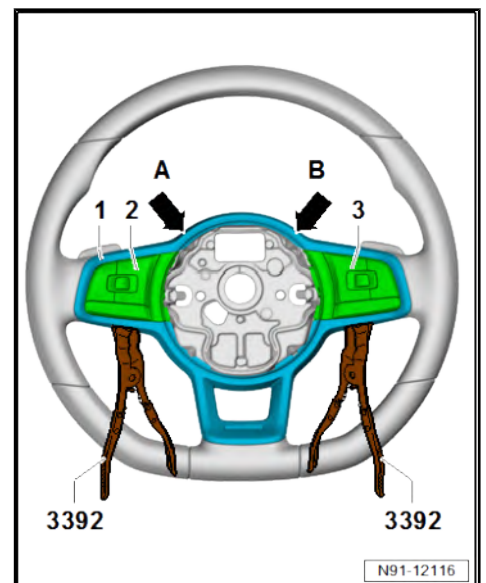
The trim is engaged in the multifunction steering wheel in areas indicated by the -arrows-.



- Using removal wedges - 3409- , release trim panel -1- in area of catches -arrows A- and -arrows B- from multifunction steering wheel.
- Using lever - T10468- , release trim panel -1- in area of multifunction buttons on left in steering wheel - E440- -2- and multifunction buttons on right in steering wheel - E441- -3-.



- Using removal pliers - 3392- , release trim -1- in area of multifunction buttons on left in steering wheel - E440- -2- and multifunction buttons on right in steering wheel - E441- -3-.
- Pull trim panel -1- in areas indicated by arrows -A- and -B- off multifunction steering wheel.
- Separate electrical connectors.
- Remove trim panel -1- together with multifunction buttons on left in steering wheel - E440- -2- and multifunction buttons on right in steering wheel - E441- -3-.





- Unscrew bolts -2- for multifunction button -1- in steering wheel on back of trim panel.
- Remove multifunction button from trim panel.

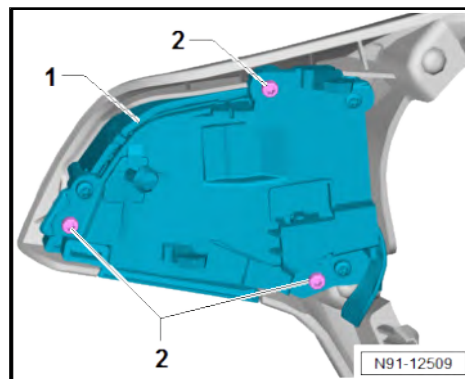
Installing

Install in the reverse order of removal, observing the following:

- Clip in trim panel from below to ease installation.

Specified torques

- ♦ ⇒ [“8.2 Assembly overview - multifunction steering wheel”, page 54](#)



8.4 Removing and installing Tiptronic switch in steering wheel -E439- / -E438-

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1783-



Note

Removal and installation is described for the left side. The removal and installation procedure on the right-hand side is basically the same.

Removing

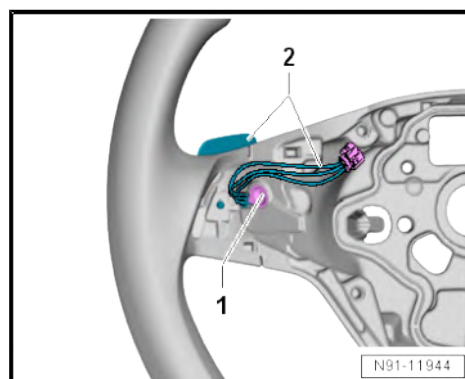
- Switch off ignition and all electrical consumers.
- Keep the ignition key outside the vehicle to prevent the ignition from being switched on unintentionally.
- Remove trim panel with multifunction buttons on right in steering wheel - E441- and multifunction buttons on left in steering wheel - E440- ⇒ [page 55](#) .
- Unscrew bolt -1- on multifunction steering wheel.
- Remove Tiptronic switch in steering wheel to shift down - E439- -2- with wire towards rear.

Installing

Install in the reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ [“8.2 Assembly overview - multifunction steering wheel”, page 54](#)





9 Connection for external multimedia devices

⇒ [“9.1 Connectors, connection for external multimedia devices”, page 59](#)

⇒ [“9.2 Removing and installing connection for external audio sources R199”, page 60](#)

9.1 Connectors, connection for external multimedia devices

⇒ [“9.1.1 Connectors, connection for external multimedia devices, pin assignments connection for external audio sources R199, USB - AUX-IN”, page 59](#)

⇒ [“9.1.2 Connectors, connection for external multimedia devices, pin assignments connection for external audio sources R199, USB socket”, page 59](#)

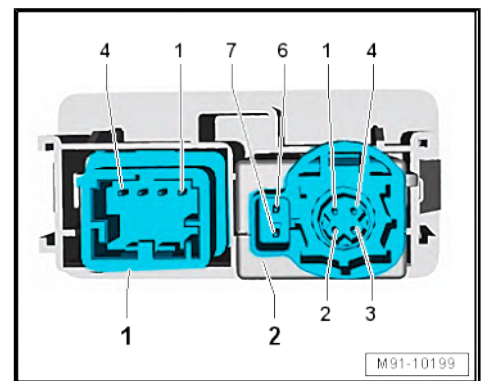
9.1.1 Connectors, connection for external multimedia devices, pin assignments connection for external audio sources - R199- , USB - AUX-IN

Electrical connector -1-, 4-pin

- 1 - Audio signal, right
- 2 - Earth
- 3 - Audio signal, left
- 4 - Earth

Connector -2-, 6-pin

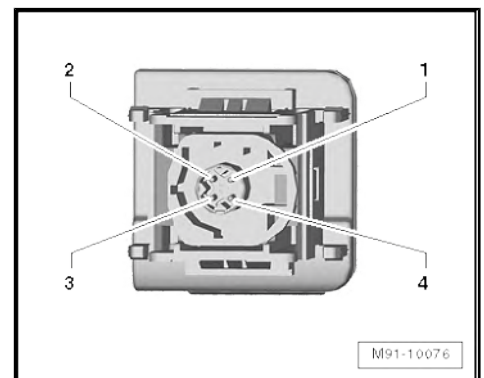
- 1 - Earth
- 2 - USB1 DATA (-)
- 3 - Not assigned
- 4 - USB1 DATA (+)
- 5 - USB +5 V
- 6 - USB earth



9.1.2 Connectors, connection for external multimedia devices, pin assignments connection for external audio sources - R199- , USB socket

Multi-pin connector, 4-pin

- 1 - DATA (+)
- 2 - VCC, +5 V
- 3 - DATA (-)
- 4 - Earth





9.2 Removing and installing connection for external audio sources - R199-

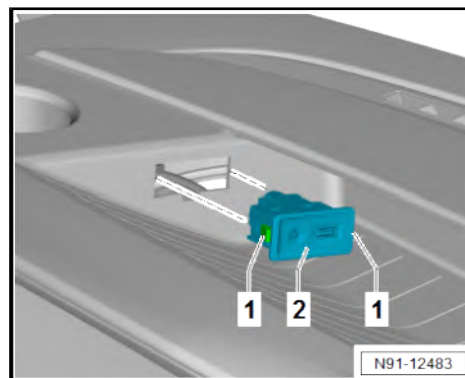
Removing

- Switch off ignition and all electrical consumers.
- Remove dash panel insert - KX2- ⇒ Electrical system; Rep. gr. 90 ; Dash panel insert; Removing and installing dash panel insert - KX2- .
- Grasp through installation slot of dash panel insert - KX2- , press clips -2-, and push connection for external audio sources - R199- -1- out of dash panel.
- Separate electrical connectors.

Installing

Install in the reverse order of removal, observing the following:

- Connection for external audio sources - R199- -1- must engage audibly in dash panel.





10 Telematics system

⇒ ["10.1 Components of telematics system", page 61](#)

10.1 Components of telematics system

The telematics system is offered as preparation for telematics.

The vehicle is prepared for this system with the following components:

- ◆ Fitting location for telematics control unit
- ◆ Electrical connector for Fleet Management Interface (FSM) in fitting location for telematics control unit
- ◆ Wiring connection with coupling point under dash panel for optional connection of telematics to tachograph (DTCO, digital tachograph)
- ◆ Right exterior mirror with telematics aerials

Vehicle must be equipped with special vehicle control unit - J608- .

The special vehicle control unit - J608- is the interface between the vehicle and the technology installed in the vehicle by third-party manufacturers. The special vehicle control unit - J608- makes information available that is needed by the telematics control unit (vehicle identification number, mileage, road speed, engine speed etc.).

Overview of preparation for telematics

1 - Right exterior mirror with GPS aerial - R50- and GSM aerial - R205-

- ☐ Removing and installing
⇒ [page 19](#)

2 - Wiring connection for GPS aerial - R50-

3 - Wiring connection for GSM aerial - R205-

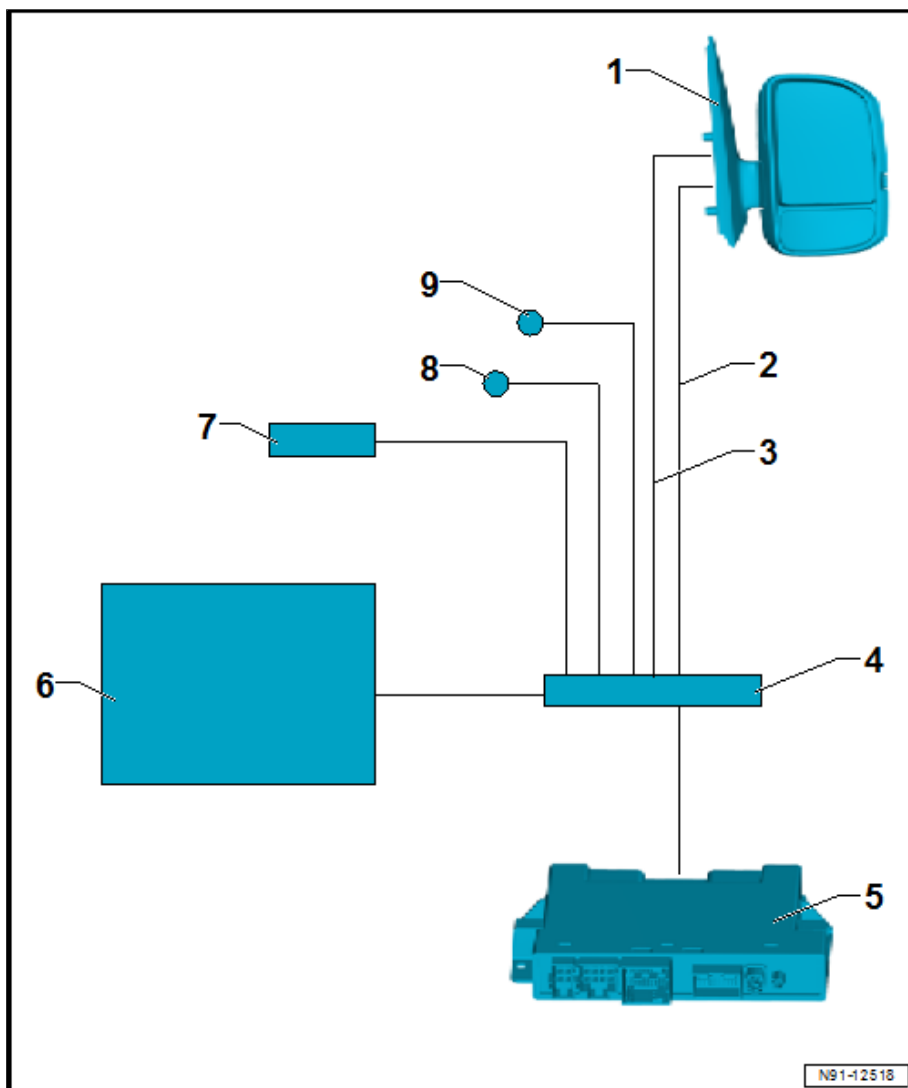
4 - Connector for Fleet Management Interface (FSM)

5 - Special vehicle control unit - J608-

- ☐ The special vehicle control unit - J608- is a prerequisite for telematics operation
- ☐ Removing and installing
⇒ Electrical system;
Rep. gr. 97 ; Control units;
Removing and installing special vehicle control unit - J608-

6 - Telematics control unit

- ☐ Not included in items supplied with preparation for telematics
- ☐ Installation of equipment from third-party manufacturers
- ☐ Fitting location, left-hand drive vehicles: behind glove compartment
- ☐ Fitting location, right-hand drive vehicles: behind glove compartment



7 - Coupling point for wiring connection from telematics control unit to tachograph

- ☐ For optional connection of telematics to tachograph
- ☐ Fitting location under auxiliary console from tachograph

8 - Power supply, terminal 15

9 - Power supply, terminal 31



11 Car-Net

⇒ [“11.1 Components of Car-Net”, page 63](#)

⇒ [“11.2 Overview of fitting locations - Car-Net services”, page 65](#)

⇒ [“11.3 Removing and installing emergency button E276”, page 66](#)

⇒ [“11.4 Removing and installing emergency call module control unit and communication unit J949”, page 67](#)

⇒ [“11.5 Commissioning ERA Glonass”, page 70](#)

11.1 Components of Car-Net

Car-Net without emergency call module control unit and communication unit - J949-

In connection with the radio navigation system Discover Media, the functions of Volkswagen Car-Net can be realised by means of the Guide & Inform Car-Net services.

In order to use the Guide & Inform Car-Net services, the vehicle must be connected to the Internet.

The internet connection can be made by tethering the customer's mobile telephone - R54- or using a USB Surf Stick offered by the vehicle manufacturer.

The customer must also be registered with the vehicle in the customer portal.

The mobile online services system is capable of self-diagnosis.

Fault finding is performed using ⇒ Vehicle diagnostic tester.

In order to deal with complaints, it is absolutely necessary to understand the functions of the mobile online services system.

For additional information refer to the ⇒ Owner's Manual , ⇒ Current flow diagrams, Electrical fault finding and Fitting locations.

Car-Net with emergency call module control unit and communication unit - J949-

In connection with the emergency call module control unit and communication unit - J949- , Car-Net services can be realised by way of emergency call + security and service.

The emergency call module control unit and communication unit - J949- is fitted in a crash-safe location behind dash panel. The emergency call module control unit and communication unit - J949- comprises a telephone module, an emergency call module and a compass module. For localisation, the emergency call module control unit and communication unit - J949- is fitted with a GPS receiver. A separate SIM card for emergency calls and roadside assistance calls is installed in the unit. Other online services are available depending on the data plan.

The mobile online services system is capable of self-diagnosis.

Fault finding is performed using ⇒ Vehicle diagnostic tester.

In order to deal with complaints, it is absolutely necessary to understand the functions of the mobile online services system.

For additional information refer to ⇒ Operating manual and ⇒ Current flow diagrams, Electrical fault finding and Fitting locations.

System overview - Car-Net without emergency call module control unit and communication unit - J949-

1 - Left exterior mirror with aerial system

- ☐ The aerial has been adapted to the vehicle's equipment
- ☐ Removing and installing ⇒ [page 24](#)

2 - Roof aerial - RX5-

- ☐ GPS aerial - R50- / GSM aerial - R205-
- ☐ Removing and installing ⇒ [page 23](#)

3 - Interior microphone - R74-

- ☐ Removing and installing ⇒ [page 35](#)

4 - Multifunction steering wheel

- ☐ Removing and installing steering wheel multi-function buttons - E441- / -E440- ⇒ [page 55](#)
- ☐ Removing and installing Tiptronic switches in steering wheel -E439- / -E438- ⇒ [page 58](#)

5 - Information electronics control unit 1 - J794-

- ☐ Removing and installing ⇒ [page 11](#)

6 - Wiring connection for infotainment CAN bus

7 - Mobile telephone - R54-

- ☐ Either a mobile telephone (WLAN Access Point) or a USB Surf Stick from the vehicle manufacturer are needed to use Car-Net.

8 - Display unit for control device of front display and information control panel - J685-

- ☐ Radio navigation system Discover Media
- ☐ Removing and installing ⇒ [page 10](#)

9 - USB Surf Stick

- ☐ Either a mobile telephone (WLAN Access Point) or a USB Surf Stick from the vehicle manufacturer are needed to use Car-Net.

10 - Connection for external audio sources - R199-

- ☐ Removing and installing ⇒ [page 60](#)

11 - Sound system

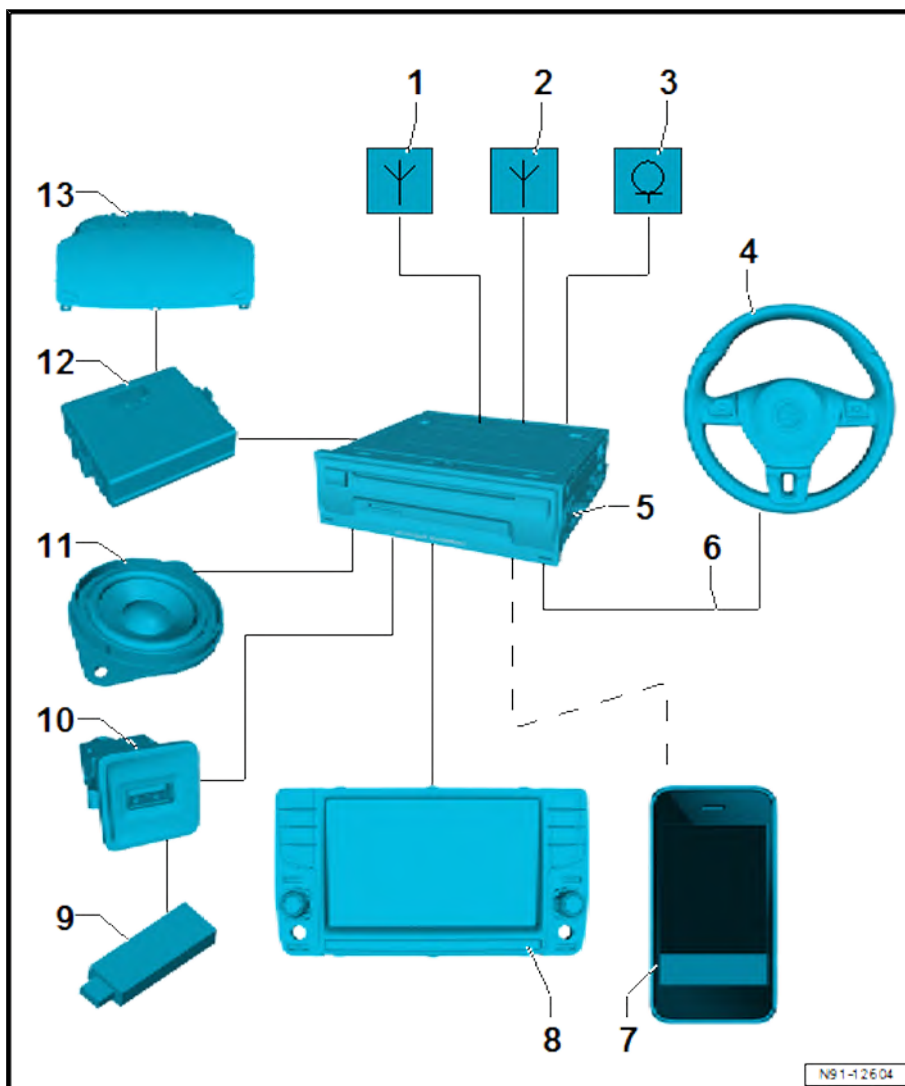
- ☐ Removing and installing ⇒ [page 13](#)

12 - Onboard supply control unit - J519-

- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 97 ; Control units; Removing and installing onboard supply control unit - J519- .

13 - Dash panel insert - KX2-

- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 90 ; Dash panel insert; Removing and installing dash panel insert - KX2-



System overview - Car-Net services with emergency call module control unit and communication unit - J949-

1 - Interior microphone - R74-

- ❑ Removing and installing
⇒ [page 35](#)

2 - Cable connection GPS aerial - R50-

- ❑ Aerial for Global Navigation Satellite System (GNSS)

3 - Roof aerial - RX5-

- ❑ GPS aerial - R50- / GSM aerial - R205-
- ❑ Removing and installing
⇒ [page 23](#)

4 - Wiring connection GSM aerial - R205-

- ❑ Telephone aerial

5 - Loudspeaker for emergency call module - R335-

- ❑ Removing and installing
⇒ [page 18](#)

6 - Emergency assistance call button - E276-

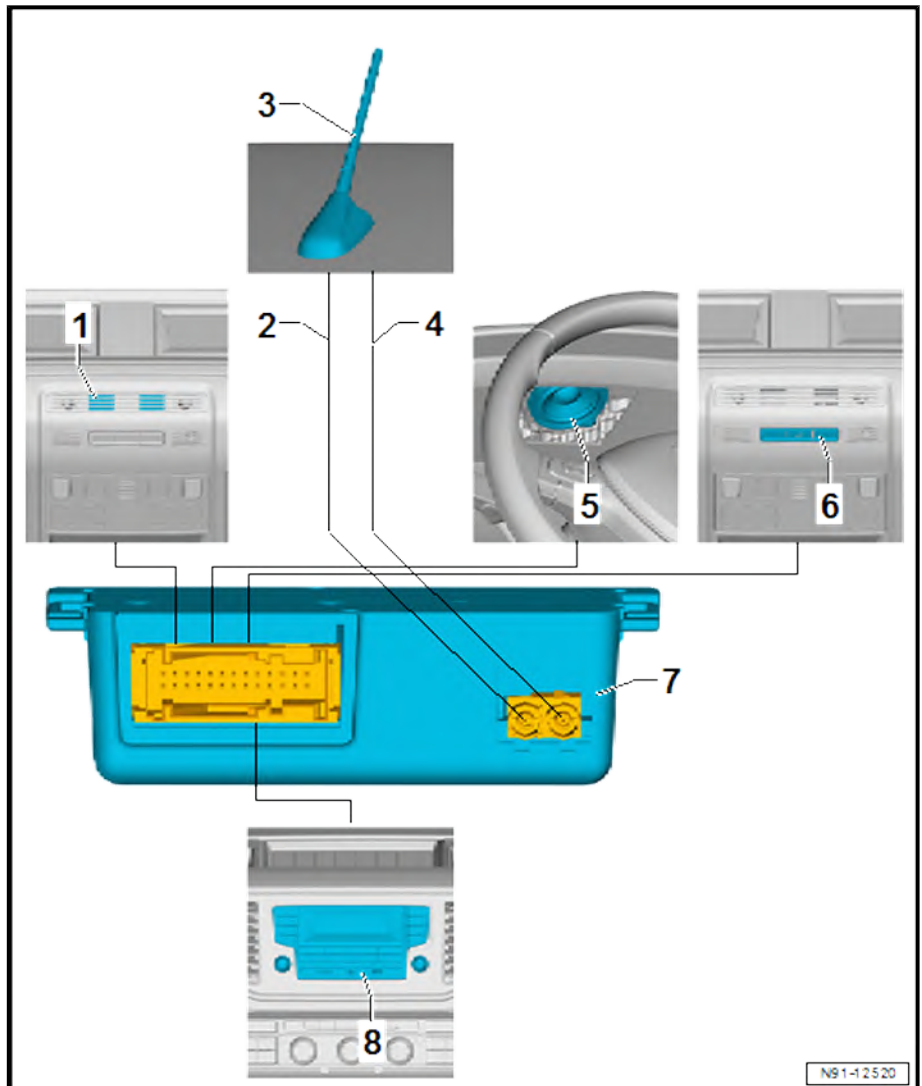
- ❑ Removing and installing
⇒ [page 66](#)

7 - Emergency call module control unit and communication unit - J949-

- ❑ Removing and installing
⇒ [page 67](#)

8 - Display unit for control device of front display and information control panel - J685-

- ❑ Removing and installing
⇒ [page 10](#)



11.2 Overview of fitting locations - Car-Net services

The overview of fitting locations for Car-Net without emergency call module control unit and communication unit - J949- is the same as the overview of fitting locations for the Infotainment system ⇒ [page 8](#) .

Overview of fitting locations - Car-Net with emergency call module control unit and communication unit - J949-



1 - Roof aerial - RX5-

- ☐ GPS aerial - R50- / GSM aerial - R205-
- ☐ Removing and installing
⇒ [page 23](#)

2 - Display unit for control device of front display and information control panel - J685-

- ☐ Removing and installing
⇒ [page 10](#)

3 - Loudspeaker for emergency call module - R335-

- ☐ Removing and installing
⇒ [page 18](#)

4 - Emergency call module control unit and communication unit - J949-

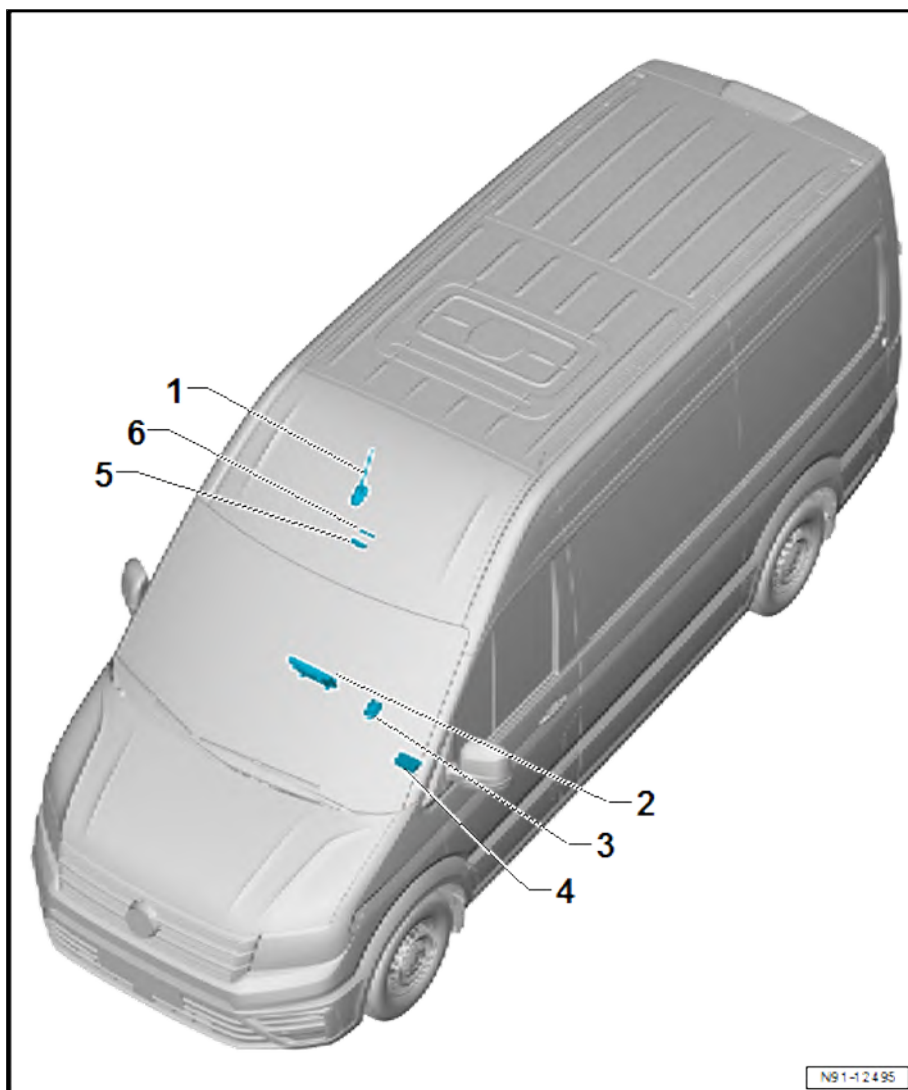
- ☐ installed above central electrical system
- ☐ Removing and installing
⇒ [page 67](#)
- ☐ Adapting ⇒ [page 68](#)

5 - Emergency assistance call button - E276-

- ☐ Integrated in Telematics button module - E734-
- ☐ Removing and installing
⇒ [page 66](#)

6 - Interior microphone - R74-

- ☐ Removing and installing
⇒ [page 35](#)



11.3 Removing and installing emergency button - E276-

Special tools and workshop equipment required

- ◆ Removal wedge - 3409- , qty. 2



Removing

- Switch off ignition and all electrical consumers.
- Remove storage compartment in roof headliner ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments and covers;



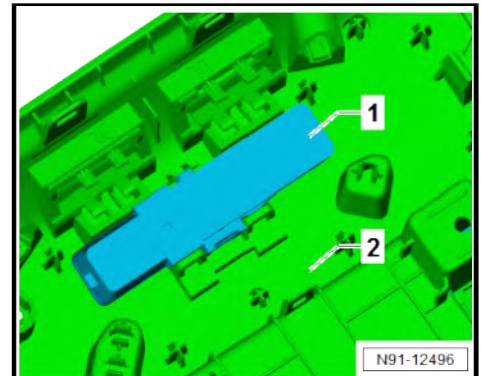
Removing and installing storage compartment in roof headliner .

- Unclip emergency assistance call button - E276- -1- from storage compartment in roof headliner -2-.
- Separate electrical connector.

Installing

Install in the reverse order of removal, observing the following:

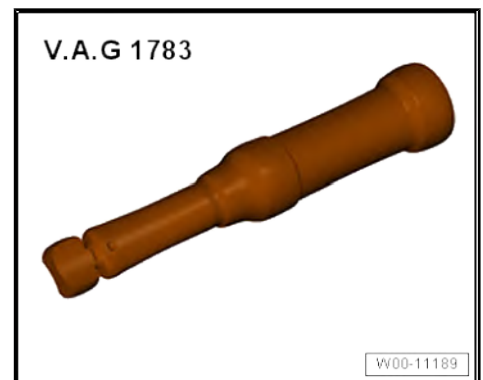
Emergency assistance call button - E276- must engage audibly during installation in storage compartment of roof headliner.



11.4 Removing and installing emergency call module control unit and communication unit - J949-

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



- ◆ Removal wedge - 3409- , qty. 2



The emergency call module control unit and communication unit - J949- is fitted above central electrical system.



Removing

- Remove dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing dash panel .
- Separate electrical connectors -4-.
- Push locking lugs -3- up and pull out emergency call module control unit and communication unit - J949- -1- in direction of -arrow- from bracket -2-.

Installing

Install in the reverse order of removal, observing the following:

If emergency call module control unit and communication unit - J949- is renewed, adapt emergency call module control unit and communication unit - J949- ➔ [page 68](#) .

Adapting control unit for emergency call module and communication unit - J949-

Special tools and workshop equipment required

- ◆ Vehicle diagnostic tester
- Data for “Guided Fault Finding” is current

Procedure

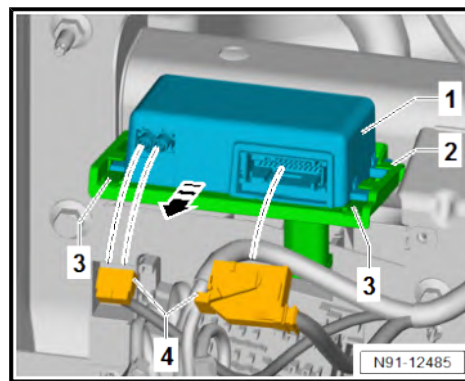
- Connect ⇒ Vehicle diagnostic tester ⇒ Maintenance ; Booklet 10.3 ; General; Connecting vehicle diagnostic tester .
- Replace control module for emergency call module and communication unit - J949- ⇒ Vehicle diagnostic tester, [Guided Functions](#), [Replacing](#).
- Remove control unit for emergency call module and communication unit - J949- ➔ [page 67](#) .
- Install new control unit for emergency call module and communication unit - J949- ➔ [page 67](#) .
- Code control module for emergency call module and communication unit - J949- ⇒ Vehicle diagnostic tester, [Guided Functions](#), [Coding](#).
- Adapt control module for emergency call module and communication unit - J949- ⇒ Vehicle diagnostic tester, [Guided Functions](#), [Adapting electronic immobilizer](#).
- Activate control module for emergency call module and communication unit - J949- ⇒ Vehicle diagnostic tester, [Guided Functions](#), [Activating](#).
- Close Guided Functions and disconnect ⇒ Vehicle diagnostic tester from vehicle.

Connect emergency call module control unit and communication unit - J949- to back end and send mileage to back end

- Drive vehicle into the open such that optimum GPS and phone reception is enabled.
- Wait for approx. 10 minutes with ignition switched on.

When the waiting time is over, check proper operation of emergency call module control unit and communication unit - J949- as follows:

- Connect ⇒ Vehicle diagnostic tester ⇒ Maintenance ; Booklet 10.3 ; General; Connecting vehicle diagnostic tester .





- Select measured values for Control module for emergency call module and communication unit - J949- ⇒ Vehicle diagnostic tester, Self-diagnosis, Reading measured values.
- Select measured values of the group for Control module for emergency call module and communication unit - J949- ⇒ Vehicle diagnostic tester, Status of activation for online services.
- The measured value for vehicle status display must be activated.

Proceed as follows if measured value for vehicle status display remains deactivated.

- Close Self-diagnosis and disconnect ⇒ Vehicle diagnostic tester from vehicle.
- Switch off ignition.
- Perform a terminal 30-type reset by disconnecting connector on emergency call module control unit and communication unit - J949- .
- Wait briefly, and reconnect connector to emergency call module control unit and communication unit - J949- .
- Connect emergency call module control unit and communication unit - J949- to back end, and send mileage to back end again ⇒ [page 68](#) .

If displayed measured value for vehicle status display is activated:

- Note mileage displayed in dash panel insert.
- Close Self-diagnosis and disconnect ⇒ Vehicle diagnostic tester from vehicle.
- Switch off ignition, and leave it switched off for at least 5 minutes.

The emergency call module control unit and communication unit - J949- connects to the back-end server, and sends the mileage to the back-end server.

Checking enrolment process in customer web portal

- Sign in to Car-Net customer portal. If necessary, create a test account for this.
- Add vehicle by entering the following:
 - ◆ Vehicle identification number
 - ◆ Mileage
 - ◆ Vehicle name



Note

- ◆ *Once the required information has been provided, a confirmation email will be sent. It is not necessary to wait for this email.*
- ◆ *After approx. 2 minutes, the vehicle is displayed under menu option My vehicles.*
- Load vehicle into dashboard to show the following values:
 - ◆ Mileage
 - ◆ Charge level



- ♦ Various functions, such as starting the charging procedure, starting the air conditioning system or locating the position where the vehicle is parked



Note

The vehicle data will be displayed only after the vehicle has been driven.

- Clear vehicle from own account.
- When customer collects vehicle, check mileage and add vehicle together with customer to customer's account.

Checking enrolment process in customer web portal

- Check proper enrolment as follows:
- Switch on ignition.
- Read measured values for Control module for emergency call module and communication unit - J949- → Vehicle diagnostic tester, Reading measured values.
- Read status of activation for Control module for emergency call module and communication unit - J949- → Vehicle diagnostic tester, Status of activation for online services.
- The displayed measured values for the subscribed Car-Net services will change to activated.

11.5 Commissioning ERA Glonass

This chapter only applies for the Russian market.

Special tools and workshop equipment required

- ♦ Vehicle diagnostic tester

Procedure



Note

ERA GLONASS cannot be commissioned until certain criteria have been fulfilled and the status LED lights up green.

If the status LED in the emergency assistance call button - E276- lights up red, proceed as follows:

- Connect ⇒ Vehicle diagnostic tester ⇒ Maintenance ; Booklet 10.3 ; General; Connecting vehicle diagnostic tester .
- Data for “Guided Fault Finding” is current
- Select emergency call module control unit and communication unit - J949- to bring ERA GLONASS into operation ⇒ Vehicle diagnostic tester, Guided Functions, Emergency call module control unit and communication unit 2, Sub-systems framework conditions, J949 Bringing ERA GLONASS into operation.

Various functions of the ERA GLONASS system are checked or executed in the program:

- ♦ Check GPS reception
- ♦ Check GSM/UMTS reception
- ♦ Check DTCs



◆ Perform terminal 30 reset

If the status LED in the emergency assistance call button - E276- still lights up red after 10 minutes, proceed as follows:

- Generate “DISS” report via “ELSA”.
- Proceed as instructed by the “NSC”.



Note

If the status LED in the emergency assistance call button - E276- lights up green and the test call fails, e.g. by redialling, submit a “DISS” note and proceed as per the instructions from the “NSC” as well.